

# A Counterexample to Kurosh's Division-Ring Problem

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## Abstract

We construct a countable division ring of characteristic zero that is algebraic over its center, generated by two elements as an algebra over that center, and infinite-dimensional over it. This gives a negative answer to the Kurosh problem for division rings.

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# 1 Introduction

A division ring  $D$  is *algebraic over its center*  $F = Z(D)$  if every  $a \in D$  satisfies a nonzero polynomial in  $F[t]$ . For a subset  $S \subseteq D$ , write  $F(S)$  for the division subring generated by  $F \cup S$ . The ring is *locally finite* if

$$[F(S) : F] < \infty \quad \text{for every finite subset } S \subseteq D.$$

The Kurosh problem for division rings asks whether algebraicity over the center implies local finiteness. We give a negative answer.

All rings and subrings in this paper have the same identity. If  $k$  is a central subfield and  $x, y$  belong to a ring,  $k[x, y]$  denotes the unital  $k$ -algebra generated by  $x, y$ ; the generators need not commute. The free polynomial algebra on two formal generators is denoted  $k\langle x, y \rangle$ .

**Theorem 1.1.** *There is a countable division ring  $D$  of characteristic zero, with center  $F = Z(D)$ , and elements  $x, y \in D$  such that*

$$D = F[x, y] = F(x, y), \quad [D : F] = \infty,$$

*while every element of  $D$  is algebraic over  $F$ .*

The same pair  $\{x, y\}$  therefore witnesses the failure of local finiteness. The construction does not prescribe the center. It first produces an algebraic division ring over a countable central subfield, and a separate finite-descent argument proves infinite dimension over the actual center. The annihilating polynomial is allowed to depend on the element.

## 1.1 Historical context

Kurosh formulated his question in 1941 in connection with Burnside's finiteness problem for groups. He asked whether a division ring that is algebraic and finitely generated as an algebra over a central subfield must be finite-dimensional over that subfield [12, Problem (K), p. 240]. The subfield in this formulation need not be the full center, and finite generation means algebra generation, not merely division generation.

The broader problem for algebraic algebras has a different history. Golod's construction, based on the Golod–Shafarevich estimates, gives finitely generated infinite-dimensional nil algebras [5, Example 1]. Their unitizations are algebraic algebras with nonzero nilpotents, so they do not answer the division-ring question.

There are positive results under additional hypotheses. Jacobson's bounded-degree theorem gives a polynomial identity, and Kaplansky's theorem makes algebraic algebras satisfying a polynomial identity locally finite; see [7, Theorems 3 and 5]. Bell, Drensky, and Sharifi later proved that a division ring whose elements satisfy nonzero polynomial relations of uniformly bounded degree  $d$ , with coefficients on the left in a possibly noncentral subfield, has dimension at most  $d^2$  over its center [2, Theorem 1.3]. Our construction imposes element-dependent relations without a uniform degree bound.

Amitsur proved that a division algebra finitely generated as an algebra over an uncountable central field is automatically algebraic [1, Corollary 2']. This conclusion is algebraicity, not local finiteness. Countability plays a different role here: it allows us to schedule all polynomial expressions in the construction. Goodearl and Letzter's recent work [6] treats nonfinite generation as an algebra for classes of transcendental skew fields and explicitly distinguishes that question

from the algebraic Kurosh problem. Theorem 1.1 concerns algebraic division rings and their actual centers.

The proof combines standard division-algebra constructions with matrix and formal-series calculations. Symbol and cyclic algebras provide division-ring models for the generic block data [4, Section 2.5]. Traces organize cyclic word relations [16, Section 4], while inner derivations describe infinitesimal conjugation [8, Theorem 8]. The lifting argument supplies the additional response, compatibility, and generation statements needed to preserve two independent deformation directions while imposing successive algebraic relations.

## 1.2 The construction

The main ingredient is Theorem 2.1. Its input is a tuple of formal series in two noncommuting variables over a finite-dimensional central division algebra. The constant tuple generates the coefficient algebra, and the two linear coefficients are independent modulo changes coming from conjugation. The theorem makes any prescribed series algebraic over a larger countable central field. At the same time it increases the degree of the division algebra, preserves all previous polynomial identities, and restores both generation and the two independent directions. These last requirements allow the construction to continue.

For the prescribed series  $w$ , the immediate goal is to find a formal substitution  $Z = (Z_1, Z_2)$  for which  $w(Z)$  satisfies the minimal polynomial of the chosen constant output  $w(Z(0))$ . The input  $Z(0)$  is a nonzero generic input, small with respect to an auxiliary valuation, not the original zero input. After splitting the coefficient algebra into matrices, we take the constant input to be a diagonal sum of independent generic matrix pairs. Varying a diagonal block gives a first eigenvalue response; adding an arrow to another block and an arrow back gives a paired response. Sections 4 and 5 establish the formal evaluation rules and show that these responses partition the eigenvalue labels into linked groups.

A transfer theorem puts each first unsolved noncommutative word coefficient in the same response space. It also turns proportional responses into functional relations between eigenvalue branches (Sections 6 and 7). The relations within each block then separate from a fixed-dimensional space of relations among blocks. To test the latter, we repeat each block and attach auxiliary blocks, called hubs. Individual hubs distinguish repeat positions; a common hub compares them. This matrix test proves that every leading error satisfies the required cyclic sums of word coefficients (Sections 8 and 9).

An eleven-block arrangement of two colors supplies corrections whose only obstructions are those cyclic sums. We solve the equations one word length at a time in Section 10. A cyclic division algebra realizes these matrix calculations over a division ring, and a Taylor translation restores generation and the two independent directions (Sections 11 and 12). The corrections are solutions of finite affine linear systems; this is what permits their descent from a splitting field to the division algebra's center. If  $w$  already satisfies its constant term's minimal polynomial, the spectral correction is unnecessary, but realization and translation are still needed to increase degree and restore the iteration hypotheses.

Section 3 first explains how the lifting theorem yields Theorem 1.1. The fields introduced during the iteration are countable, so every polynomial expression can be scheduled. A division-ring ultraproduct retains the resulting relations. Its two-generated subalgebra is already a division ring, and finite word sets prove that its dimension over the central coefficient field is infinite. The actual-center argument concludes this deduction. The remaining sections prove the lifting theorem and all its required compatibilities.

## 2 The lifting theorem

The construction proceeds through finite-dimensional central division algebras. At each step we impose an algebraic relation on one prescribed element while preserving the freedom to enlarge the algebra at the next step. Formal power series in two noncommuting variables record that freedom.

For a  $k$ -algebra  $A$ , write

$$A\langle\langle z_1, z_2 \rangle\rangle$$

for the ring of formal series in words in  $z_1, z_2$ , with coefficients in  $A$  commuting with the variables. Multiplication concatenates words and multiplies their coefficients in their original order. The ring is complete for the filtration by word length. For a tuple  $X = (X^{(1)}, \dots, X^{(u)})$  of such series, let  $X_0 \in A^u$  be its constant tuple and let  $X_1, X_2 \in A^u$  be its coefficients of the two individual letters. Thus the subscripts 1, 2 denote letters, not degrees. Set

$$J_A(X_0) = \{([T, X_0^{(1)}], \dots, [T, X_0^{(u)}]) : T \in A\}, \quad [T, a] = Ta - aT. \quad (2.1)$$

The classes of  $X_1, X_2$  modulo this subspace describe changes of the tuple that cannot be obtained by infinitesimal conjugation.

**Theorem 2.1** (Algebraic lifting). *Let  $k$  be a countable field of characteristic zero containing every root of unity, and let  $A$  be a finite-dimensional central division algebra over  $k$ . Suppose  $X \in A\langle\langle z_1, z_2 \rangle\rangle^u$  satisfies:*

- (i)  $X_0$  generates  $A$  as a  $k$ -algebra;
- (ii)  $X_1, X_2$  have linearly independent classes in  $A^u/J_A(X_0)$ .

*For every  $w \in A\langle\langle z_1, z_2 \rangle\rangle$ , there exist a countable field extension  $k'/k$ , a central division algebra  $A'/k'$  of strictly larger degree, and series*

$$X' \in A'\langle\langle s_1, s_2 \rangle\rangle^u, \quad w' \in A'\langle\langle s_1, s_2 \rangle\rangle$$

*with the following properties:*

- (a) *Every noncommutative polynomial identity over  $k$  satisfied by the tuple  $(X, w)$  is satisfied by  $(X', w')$  as an identity of formal series.*
- (b) *The constant tuple  $X'_0$  generates  $A'$  over  $k'$ , and the classes of  $X'_1, X'_2$  in  $(A')^u/J_{A'}(X'_0)$  are linearly independent.*
- (c) *There is a monic nonzero polynomial  $q \in k'[t]$  such that  $q(w') = 0$ .*

*Here the degree of a central division algebra  $B/F$  is  $\sqrt{[B : F]}$ .*

Theorem 2.1 is the central construction. Its application to the Kurosh problem uses  $u = 2$  and addresses one polynomial in the two generators at a time. The fields are allowed to grow; they need not be algebraic extensions of their predecessors. The final step will therefore identify the actual center separately and prove infinite dimension over it.

Figure 1 separates the two tasks within a lifting step. The response calculation makes the prescribed series algebraic; the final translation restores the hypotheses needed to repeat the step. The two transverse directions must survive together, not merely one at a time.

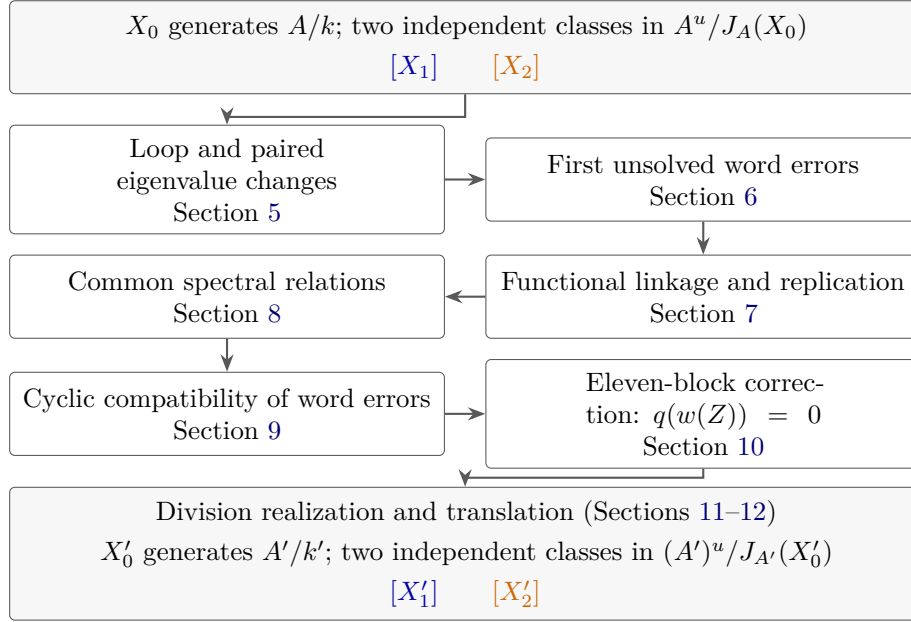


Figure 1: The spectral case of Theorem 2.1. The two displayed classes at each end are linearly independent in the indicated quotient. The middle boxes describe the matrix calculation: allowable eigenvalue changes control the first unsolved word errors, and the cyclic relations are precisely the obstructions to correction. Realization and translation return a generating tuple over a division algebra of strictly larger degree. If  $w$  already satisfies the minimal polynomial of  $w(0)$ , the middle spectral correction is omitted, but realization and translation remain necessary. No embedding tower of the successive division algebras is asserted.

### 3 The countable construction and the actual center

We now apply Theorem 2.1 repeatedly. Recall its full conclusion: from a finite-dimensional central division algebra  $A/k$  and a tuple of free series  $X$  whose constant term generates  $A$  and whose two linear coefficients are independent modulo inner directions, it produces such a tuple over a countable extension  $k'$  and a central division algebra of strictly larger degree. It preserves every old polynomial identity and gives a monic algebraic relation for any chosen series  $w$ . The countability, generation, and two-direction conclusions are all needed to iterate this theorem.

The resulting division algebras need not embed in one another. We use an ultraproduct to retain their polynomial relations, and then prove separately that the resulting algebra has infinite dimension over its actual center.

#### 3.1 Scheduling all polynomial expressions

Start with

$$k_0 = \overline{\mathbb{Q}}, \quad A_0 = k_0, \quad X^{(0)} = (z_1, z_2).$$

Its constant pair  $(0, 0)$  generates  $k_0$  as a unital  $k_0$ -algebra. The inner-direction space is zero, and its two linear coefficients  $(1, 0)$  and  $(0, 1)$  are independent. Thus it satisfies the hypotheses of Theorem 2.1.

Inductively we obtain countable fields

$$k_0 \subseteq k_1 \subseteq k_2 \subseteq \cdots,$$

finite-dimensional central division algebras  $A_i/k_i$  of strictly increasing degrees, and pairs

$$X^{(i)} \in A_i \langle\langle z_1, z_2 \rangle\rangle^2$$

satisfying the same generation and transverse-independence hypotheses. Write  $(x_i, y_i) = X_0^{(i)}$  for the constant pair; in particular,

$$A_i = k_i[x_i, y_i]. \tag{3.1}$$

Here is a schedule that ensures algebraicity of every expression needed in the limit. When  $k_i$  has been constructed, enumerate the countable set  $k_i \langle x, y \rangle$  as  $(p_{i,j})_{j \geq 0}$ . Enumerate the pairs  $(i, j) \in \mathbb{N}^2$  by increasing  $i + j$ , in a fixed order within each diagonal. If  $(i_n, j_n)$  is the  $n$ th pair, beginning at  $n = 0$ , then  $i_n \leq n$ . At stage  $n$ , apply the lifting theorem to

$$w = p_{i_n, j_n}(X^{(n)}).$$

This expression is defined over  $k_n$ , since  $k_{i_n} \subseteq k_n$ . The identity  $w - p_{i_n, j_n}(X^{(n)}) = 0$  is preserved, so the new relation for the lifted  $w$  gives a nonzero monic polynomial

$$q_{i_n, j_n}(t) \in k_{n+1}[t]$$

such that

$$q_{i_n, j_n}(p_{i_n, j_n}(X^{(n+1)})) = 0. \tag{3.2}$$

Every later lifting preserves this polynomial identity, now regarded over its coefficient field  $k_{n+1}$ . Taking constant terms shows that the corresponding identity holds at  $(x_h, y_h)$  for every  $h \geq n + 1$ . Each polynomial over each  $k_i$  is treated at a finite stage.

### 3.2 A two-generated algebraic division ring

Fix a nonprincipal ultrafilter  $\mathcal{U}$  on  $\mathbb{N}$  and form (see [10] for the general construction)

$$\mathcal{D} = \prod_{i \geq 0} A_i / \mathcal{U}.$$

Here two sequences represent the same element when their equality set belongs to  $\mathcal{U}$ . This is a division ring: a nonzero class has nonzero coordinates on a set in  $\mathcal{U}$ , and taking their coordinatewise inverses on that set gives its inverse.

Put  $k_\infty = \bigcup_i k_i$ . An element  $c \in k_j$  determines a class in  $\mathcal{D}$  by taking its image in the central field  $k_i \subset A_i$  for  $i \geq j$  and choosing arbitrary coordinates at the finitely many earlier indices. These classes are independent of the early choices and of  $j$ . They define a central field embedding

$$k_\infty \hookrightarrow \mathcal{D} :$$

the field operations agree at every sufficiently large index, and a nonzero field element stays nonzero at all those indices. This construction uses only the inclusions of coefficient fields; it does not require embeddings between the algebras  $A_i$ .

Let

$$x = [(x_i)_i], \quad y = [(y_i)_i], \quad R = k_\infty[x, y] \subseteq \mathcal{D}.$$

Every  $r \in R$  has the form  $p(x, y)$  for some  $p \in k_j\langle x, y \rangle$ . The schedule and (3.2) provide a nonzero monic polynomial  $q \in k_h[t]$ , for a suitable  $h$ , such that  $q(p(x_i, y_i)) = 0$  at every sufficiently large index. Therefore  $q(r) = 0$  in  $\mathcal{D}$ . Every element of  $R$  is algebraic over  $k_\infty$ .

This already makes  $R$  a division ring. For  $0 \neq r \in R$ , choose a nonzero relation of least degree

$$a_0 + a_1 r + \cdots + a_n r^n = 0, \quad a_j \in k_\infty.$$

Its constant coefficient is nonzero: otherwise multiplication by  $r^{-1}$  in the ambient division ring  $\mathcal{D}$  would give a shorter relation. Consequently

$$r^{-1} = -a_0^{-1}(a_1 + a_2 r + \cdots + a_n r^{n-1}) \in R.$$

Since  $k_\infty$  is countable and  $R$  is generated by two elements as an algebra,  $R$  is countable. Its characteristic is zero.

### 3.3 Infinite dimension and passage to the center

We first prove

$$[R : k_\infty] = \infty. \tag{3.3}$$

For each  $i$  and  $\ell \geq 0$ , let  $V_{i,\ell}$  be the  $k_i$ -span of the words in  $x_i, y_i$  of length at most  $\ell$ , including the empty word 1. If  $V_{i,\ell} = V_{i,\ell+1}$ , then  $V_{i,\ell}$  is stable under right multiplication by  $x_i$  and  $y_i$  and contains every word. By (3.1), it equals  $A_i$ . Until this happens, the dimensions increase strictly, starting from 1.

Fix  $\ell$ . The dimensions  $[A_i : k_i]$  tend to infinity, so

$$\dim_{k_i} V_{i,\ell} \geq \ell + 1$$

for every sufficiently large  $i$ . There are only finitely many subsets of size  $\ell + 1$  in the finite list of words of length at most  $\ell$ . The ultrafilter therefore selects one fixed subset  $w_0, \dots, w_\ell$  that is linearly independent at a set of stages belonging to  $\mathcal{U}$ .

These same words are independent over the embedded  $k_\infty$  in  $R$ . Indeed, the coefficients of a putative relation

$$\sum_{j=0}^{\ell} c_j w_j(x, y) = 0$$

all belong to one field  $k_h$ . Equality in the ultraproduct holds on a set in  $\mathcal{U}$ . Intersecting it with the independence set and the tail  $i \geq h$  gives a nonempty set where stage independence forces every  $c_j$  to vanish. Thus  $R$  has arbitrarily large finite linearly independent sets, proving (3.3).

Finally let  $F = Z(R)$  be the actual center. It contains  $k_\infty$  and is algebraic over  $k_\infty$ , since every element of  $R$  is. In particular, every element of  $R$  is algebraic over  $F$ . Also

$$R = k_\infty[x, y] = F[x, y] = F(x, y),$$

because  $F \subseteq R$  and  $R$  is already a division ring.

We use finite descent of the multiplication table and generator coordinates; compare the proof of the local-finiteness criterion in [3, Section 4]. Suppose, for a contradiction, that  $[R : F] = n < \infty$ . Choose an  $F$ -basis  $b_1 = 1, b_2, \dots, b_n$  of  $R$ , and write

$$b_i b_j = \sum_{h=1}^n c_{ijh} b_h, \quad x = \sum_{h=1}^n u_h b_h, \quad y = \sum_{h=1}^n v_h b_h,$$

with all coefficients in  $F$ . Let  $E$  be the subfield of  $F$  generated over  $k_\infty$  by these finitely many coefficients. Algebraicity gives  $[E : k_\infty] < \infty$ . The space

$$B = \sum_{h=1}^n E b_h$$

contains  $1, x, y$  and is closed under multiplication by the displayed structure constants. It is therefore a  $k_\infty$ -subalgebra of  $R$  containing  $k_\infty[x, y] = R$ , so  $B = R$ . This implies

$$[R : k_\infty] \leq n[E : k_\infty] < \infty,$$

contrary to (3.3).

Thus the countable characteristic-zero division ring  $D = R$  is algebraic over its actual center  $F$ , while

$$D = F[x, y] = F(x, y), \quad [D : F] = \infty.$$

The finite set  $\{x, y\}$  witnesses the failure of local finiteness.

This proves Theorem 1.1, subject only to Theorem 2.1, which we now prove.

## 4 Formal evaluations and generic spectra

We begin the proof of Theorem 2.1 by constructing a substitution  $Z = (Z_1, Z_2)$  for the two free letters. Its coefficients will commute with the old algebra  $A$ , so applying it to  $(X, w)$  preserves their polynomial identities. The matrix calculation focuses on  $w$ ; Sections 11 and 12 realize the substitution in a division algebra of larger degree and restore generation and both transverse directions for  $X(Z)$ .

Throughout the lifting argument,  $k$  is a field of characteristic zero containing all roots of unity,  $A$  is a central division algebra of degree  $d$  over  $k$ , and

$$w \in A \langle\langle z_1, z_2 \rangle\rangle.$$



The letters commute with  $A$ , but not with each other. Thus an element is a family  $(a_\omega)_\omega$  indexed by finite words, written  $\sum_\omega a_\omega z^\omega$ , and multiplication uses concatenation of words. Fix an algebraic closure  $K$  of  $k$  and a splitting  $A \otimes_k K \simeq M_d(K)$ ; this uses the splitting and scalar-extension properties of central simple algebras [14, Chapter IV, Example 2.14(a) and Proposition 2.15].

If  $P = (P_1, P_2)$  is a pair of  $n \times n$  indeterminate matrices, put

$$W_n(P) = \sum_\omega a_\omega \otimes P^\omega \in M_{dn}(K[[P]]).$$

Here  $K[[P]]$  is the ordinary commutative power series ring in the  $2n^2$  entries. We omit the subscript  $n$  when the size is clear. The coefficient matrices act on the first tensor factor. In particular, every input arrow between blocks is a pair of scalar matrices acting on the second factor. It is not an arbitrary pair of matrices on both factors.

In the spectral case we seek a substitution  $Z = (Z_1, Z_2)$ , with constant term  $R$ , for which  $W(Z)$  is conjugate to a split semisimple constant output  $W(R)$ . If  $\Lambda$  is the set of distinct eigenvalues of  $W(R)$ , this gives

$$q(W(Z)) = 0, \quad q(t) = \prod_{\lambda \in \Lambda} (t - \lambda).$$

The division realization will identify  $q$  with the minimal polynomial over its central coefficient field. If  $w$  already satisfies the minimal polynomial of  $w(0)$ , substitution retains that relation and the spectral correction is unnecessary.

For the spectral construction, we will find matrix sizes with a common eigenvalue multiplicity, disjoint spectra at independent inputs, and scalar spectral blocks under commuting Taylor perturbations. These properties will make the eigenvalue responses of Section 5 well-defined. We first specify the evaluations and specializations used to establish them and to evaluate triangular auxiliary inputs later.

## 4.1 Convergence and specialization

Let  $E$  be a complete discretely valued field containing  $K$ , with valuation  $v$ , such that  $v(c) = 0$  for  $c \in K^\times$ . We use  $E = C((\varepsilon))$ , or a finite extension of such a field, and normalize  $v(\varepsilon) = 1$ . Matrix valuations mean the minimum of the entry valuations. A matrix is *small* if this minimum is positive. A small generic point is  $P = \varepsilon P^\circ$ , where all entries of  $P^\circ$  are algebraically independent over the previously specified constant field. Several independent points can use the same  $\varepsilon$ .

The finite extensions used to split spectra are separable in characteristic zero. The valuation extends to each such extension, which remains complete and discretely valued [13, Theorem 7.38 and Remark 7.43]; with the normalization  $v(\varepsilon) = 1$ , its value group may be  $\frac{1}{e}\mathbb{Z}$  for a positive integer  $e$ .

**Lemma 4.1** (Admissible evaluations). *Suppose a constant input pair  $R$  is block upper triangular with  $b$  blocks and small diagonal blocks. Its strictly upper blocks may have arbitrary entries in  $E$ . Let  $H$  be a matrix pair of formal series of positive order in finitely many commuting or free variables. Then  $W(R + H)$  is defined coefficientwise by convergent sums in  $E$ . The same assertion holds for products, finite determinants, and derivatives of fixed order of finitely many such entry series.*

*These evaluations preserve formal entry identities. They also remain valid coefficientwise in an additional parameter  $\eta$  if the diagonal blocks have entries in  $E[[\eta]]$  with small constant terms, while the strictly upper blocks and the coefficients of  $H$  have entries in  $E((\eta))$ .*

*Proof.* Fix a coefficient of degree  $r$  in the variables of  $H$ . An input path contributing to this coefficient uses at most  $r$  positive-order terms of  $H$ . Between these insertions it follows the constant upper triangular pair. In each of the at most  $r + 1$  resulting intervals, there are at most  $b - 1$  strictly upper steps. Thus at most  $(r + 1)(b - 1)$  constant factors are strictly upper. All remaining constant factors come from small diagonal blocks.

Only finitely many coefficients of  $H$  can occur at the chosen degree. Their valuations, and those of the finitely many strictly upper blocks, have a common lower bound. If the original word has length  $L$ , its contribution consequently has valuation at least  $cL - C_r$ , where  $c > 0$  and  $C_r$  is independent of the word. There are finitely many words of each length, so the sum converges. The same estimate applies after inserting the finitely many extra factors needed for a product or a derivative of fixed order. Completeness allows regrouping of these sums, proving compatibility with multiplication and with identities.

For the last assertion, first fix both the degree  $r$  in  $H$  and an  $\eta$ -coefficient. The bounded number of strictly upper steps and of  $H$ -insertions bounds their total negative  $\eta$ -order. Hence only a bounded number of positive-order diagonal  $\eta$ -terms can contribute, and only finitely many of their coefficients are used. Every other factor is a small constant diagonal term. The preceding valuation estimate therefore applies again. The resulting assertion is coefficientwise; no interchange of the two Laurent topologies is being asserted.  $\square$

We will use inverses of finite matrices as rational operations after evaluation. The following elementary distinction is important. Although  $(\varepsilon - Q)^{-1}$  has a formal expansion at  $Q = 0$  over  $E$ , that expansion need not converge after  $Q = \varepsilon Q^\circ$ . Accordingly, when transferring an identity involving an inverse, we first multiply by its determinant. We evaluate the resulting numerator by Lemma 4.1, and divide only when the evaluated determinant is nonzero. Formal inverse expansions are used to define coefficients; they are not presumed to converge at every subsequent specialization.

**Lemma 4.2** (Detection by independent residues). *Let*

$$F(Q) = \sum_{\alpha} c_{\alpha} Q^{\alpha} \in E[[Q_1, \dots, Q_N]], \quad \inf_{\alpha} v(c_{\alpha}) > -\infty.$$

*Adjoin elements  $Q_1^\circ, \dots, Q_N^\circ$  whose residues are algebraically independent over the residue field of  $E$ , and complete with respect to the extended valuation. Then  $F(\varepsilon Q^\circ)$  is defined, and is nonzero whenever  $F$  is nonzero.*

*Proof.* The valuations  $v(c_{\alpha}) + |\alpha|$  tend uniformly to infinity with  $|\alpha|$ . Their least value is therefore attained by a nonempty finite set of monomials. After scaling to this valuation, the residue of their sum is a nonzero polynomial in the independent residues of the  $Q_j^\circ$ . This proves both assertions.  $\square$

In applications, the coefficients of a cleared numerator are finite sums of products of base-point scalars and matrix-entry series. The path estimate gives a uniform valuation bound in the auxiliary coordinates. Lemma 4.2 therefore identifies formal identities in these coordinates with identities at independent small generic points. The base-point valued field is constructed first; the auxiliary residues are adjoined afterward. This order keeps their independence valid even after the base spectrum has been split. The argument can be applied separately at each Taylor degree.

We record two further field conventions. Evaluation

$$K[[P]] \longrightarrow C(P^\circ)((\varepsilon)), \quad P \longmapsto \varepsilon P^\circ,$$

is injective, by homogeneous degree or by Lemma 4.2. It extends to the fraction field, and a finite splitting field embeds into a finite algebraic extension of the target. Also, if  $R$  is a small constant pair, then

$$K[[P]] \longrightarrow E[[\Delta]], \quad P \longmapsto R + \Delta \quad (4.1)$$

is injective: reduction modulo the valuation ideal sends the image of  $f$  to  $f(\Delta)$ . We use this observation for small block diagonal  $R$ . Denominators in the fraction field remain nonzero, so the map extends to fraction fields.

Finally, the Taylor substitution  $P \mapsto P + \eta v$  maps  $\text{Frac } K[[P]]$  into  $\text{Frac } K[[P]][[\eta]]$ , with reduction the identity. It extends to a finite separable extension  $\Omega$  with values in  $\Omega[[\eta]]$ : choose a primitive element and lift its defining equation using its nonzero derivative. The lift reducing to the chosen primitive element is unique, by coefficient recursion. This convention lets us move a base point while retaining its labeled eigenvalue branches.

## 4.2 Generic division evaluations

Generic matrix evaluations have a useful divisibility property for their eigenvalue multiplicities. We obtain it by first evaluating inside a division algebra.

For  $n \geq 1$ , choose independent central variables  $\alpha, \beta$  and a primitive  $n$ th root of unity  $\zeta_n \in k$ . Over  $k_0 = k(\alpha, \beta)$  let  $B_n$  be generated by  $x, y$  with

$$x^n = \alpha, \quad y^n = \beta, \quad yx = \zeta_n xy.$$

For  $n = 1$  this means  $B_1 = k_0$ . The reduced monomials  $x^i y^j$ ,  $0 \leq i, j < n$ , form a basis. Indeed they span by the relations, and after adjoining  $n$ th roots of  $\alpha$  and  $\beta$  their clock-and-shift matrices are linearly independent. These matrices span  $M_n$ , so  $B_n$  is central simple of degree  $n$ . This is the standard symbol construction [4, Corollary 2.5.5(1)], with the inverse root of unity for our convention.

**Lemma 4.3.** *The central simple algebra  $A \otimes_k B_n$  is division. Its pure rational scalar extensions and its central Laurent series extensions are division as well.*

*Proof.* Start with  $A((x))$ , where  $x$  commutes with  $A$ , and let  $\sigma$  fix  $A$  and send  $x$  to  $\zeta_n x$ . The skew Laurent series ring  $A((x))((y; \sigma))$ , with  $ya = \sigma(a)y$ , is a division ring by the leading-coefficient construction of inverses. The elements  $x^n, y^n$  are central and algebraically independent. The defining relations give a unital map of  $A \otimes_k B_n$  into this division ring. Central simplicity makes the map injective. A finite dimensional domain over a field is division, since multiplication by a nonzero element is an injective and hence surjective linear map.

For a central division algebra  $D/F$ , the rational extension  $D(t)$  embeds in the division ring  $D((t))$ . Iterating proves the rational assertion. Because  $D$  has a finite  $F$ -basis,  $D \otimes_F F((\varepsilon)) = D((\varepsilon))$ , which is again division by leading coefficients.  $\square$

Choose a basis  $b_1, \dots, b_{n^2}$  of  $B_n$ , and let  $c_{ij}$ ,  $i = 1, 2$ ,  $j = 1, \dots, n^2$ , be independent variables. Evaluate  $w$  at

$$P_i = \varepsilon \sum_j c_{ij} b_j.$$

This gives an element of a central division algebra of degree  $dn$  over

$$F_n = k(\alpha, \beta, (c_{ij}))((\varepsilon)).$$

After splitting  $A$  and  $B_n$  by constants independent of the  $c_{ij}$ , the  $b_j$  become a basis of  $M_n$ . The change from the  $c_{ij}$  to the entries of  $P_i/\varepsilon$  is therefore an invertible linear change of independent coordinates. The resulting evaluation is a small generic scalar matrix evaluation in the sense fixed above.

Let  $e_n$  denote the minimal-polynomial degree of this element. Its minimal polynomial is irreducible and separable over  $F_n$ . Consequently  $W(P)$  is semisimple over a splitting field, and every distinct root has the same multiplicity

$$t_n = \frac{dn}{e_n}. \quad (4.2)$$

For completeness, if  $D/F$  has degree  $N$  and  $a \in D$  has minimal polynomial  $q$  of degree  $e$ , then left multiplication by  $a$  has characteristic polynomial  $q^{N^2/e}$ : view  $D$  as a vector space over  $F[a]$ . After splitting  $D = M_N$ , left multiplication acts by the matrix  $a$  on each of its  $N$  columns. Its characteristic polynomial is therefore the  $N$ th power of the matrix characteristic polynomial. Comparing root multiplicities gives  $N/e$  in the matrix representation. This also proves that  $N/e$  is an integer.

Linear independence of  $1, a, \dots, a^{e-1}$  is preserved by field extension. Thus the same  $e_n, t_n$  describe  $W(P)$  over a splitting field of  $\text{Frac } K[[P]]$ .

### 4.3 Nonconstant roots and stable multiplicity

Let  $q_0$  be the minimal polynomial of  $w(0)$  over  $k$ . If  $q_0(w) = 0$  as a free series, the subsequent spectral construction is unnecessary; the unconstrained case of the lifting argument will use this existing relation. In the rest of this section assume  $q_0(w) \neq 0$ .

A nonzero homogeneous part of  $q_0(w)$  is detected by scalar matrices of every sufficiently large size. To see this, choose a word of length  $L$  with nonzero coefficient and use a path of  $L + 1$  vertices, placing its successive letters on the consecutive matrix-unit arrows. The  $(1, L + 1)$  entry detects precisely that coefficient. Padding by zero blocks works for all larger sizes. It follows that  $q_0(W(P)) \neq 0$  at every generic size  $n \geq n_0$ , for some  $n_0$ .

**Lemma 4.4.** *For  $n \geq n_0$ , every distinct generic eigenvalue of  $W(P)$  has a nonzero differential in the input entries. Independent small generic points of sizes at least  $n_0$  have disjoint spectra. Their spectra also avoid the constant spectrum of  $w(0)$ .*

*Proof.* Use the division evaluation over  $F_n$ . Each derivation  $\partial/\partial c_{ij}$  acts coefficientwise on Laurent series and extends uniquely to algebraic extensions in characteristic zero. Uniqueness implies that it commutes with automorphisms of a splitting field over  $F_n$ . If all these derivations killed one root, they would kill all its Galois conjugates. Irreducibility makes these all the distinct roots. Thus every coefficient of the reduced characteristic polynomial would have all its input derivatives zero.

The coefficient of  $\varepsilon^r$  in such a characteristic coefficient is a homogeneous polynomial of degree  $r$  in the  $c_{ij}$ . In characteristic zero a polynomial whose partial derivatives all vanish is constant. All terms of positive  $\varepsilon$ -degree therefore vanish. The characteristic polynomial is the characteristic polynomial of  $w(0)$ . Semisimplicity then implies  $q_0(W(P)) = 0$ , a contradiction.

Splitting  $A, B_n$  uses constants independent of the inputs. The invertible basis change described above, together with the nonzero factor  $\varepsilon$ , transports the differential assertion to ordinary matrix-entry coordinates. For independent points  $P, Q$ , derivations in the  $P$ -coordinates kill the roots belonging to  $Q$ . They cannot kill a root belonging to  $P$ . No root can therefore belong to both spectra. The same argument excludes a constant root.  $\square$

Choose  $p \geq n_0$  for which the positive integer  $t_p$  is minimal among all  $t_n$ ,  $n \geq n_0$ , and put

$$t = t_p, \quad \mathcal{S} = \{pN : N \geq 1\}.$$

**Lemma 4.5.** *For every  $N \geq 1$ ,*

$$t_{pN} = t, \quad e_{pN} = Ne_p.$$

*Proof.* Specialize a size- $pN$  point to a diagonal sum of  $N$  independent size- $p$  points. Lemma 4.4 gives  $Ne_p$  distinct eigenvalues there. The generic bound on minimal-polynomial degree survives this specialization: it is expressed by the vanishing of all  $(e_{pN} + 1)$ -minors of the matrix whose columns are the vectorizations of  $1, W, \dots, W^{e_{pN}}$ . These are formal entry identities covered by Lemma 4.1. Consequently  $e_{pN} \geq Ne_p$ . Minimality of  $t$  gives the opposite inequality,  $e_{pN} = dpN/t_{pN} \leq dpN/t = Ne_p$ .  $\square$

We henceforth use sizes in  $\mathcal{S}$ . At size  $n$ , the minimal polynomial always has degree at most  $e_n = dn/t$  at every admissible specialization. The following consequence controls all commuting Taylor deformations used below.

**Lemma 4.6** (Scalar branches at maximal root count). *Let  $V(T)$  be a matrix over a commuting formal Taylor ring  $E[[T]]$ . Suppose  $V(0)$  is split semisimple with  $e$  distinct eigenvalues and the minimal-polynomial degree of  $V(T)$  over  $\text{Frac } E[[T]]$  is at most  $e$ . Then a formal change of basis reducing to the identity separates  $V(T)$  into  $e$  scalar blocks. The sizes of these blocks are their multiplicities at  $T = 0$ .*

*Proof.* Use the eigenspace decomposition at zero. Off-diagonal blocks can be removed recursively in total Taylor degree: the equation for the block between eigenvalues  $\lambda_i, \lambda_j$  is multiplication by  $\lambda_i - \lambda_j$ , a nonzero scalar. We obtain blocks  $V_i(T)$  reducing to  $\lambda_i I$ . Their characteristic polynomials have pairwise unit resultants, because their reductions have different roots. Their minimal polynomials over the Taylor fraction field are therefore pairwise coprime. The minimal-polynomial degree of their direct sum is the sum of their degrees. There are  $e$  nonempty blocks and the total degree is at most  $e$ , so every block has minimal-polynomial degree one. Each  $V_i(T)$  is scalar. Its scalar entry belongs to  $E[[T]]$ , since the block entries do.  $\square$

In particular, the roots at a small semisimple point with  $e_n$  distinct roots have labeled scalar Taylor branches. The splitting field of the generic point embeds in their Taylor fraction field by (4.1); a labeling is chosen by reduction to the roots at the base.

Let  $\rho_n$  be the generic rank of the input Jacobian of

$$\text{tr } W, \text{tr } W^2, \dots, \text{tr } W^{dn}.$$

Its minors are formal entry identities, so  $\rho_n$  bounds the corresponding specialized ranks. At a maximal-root-count point the scalar branches satisfy

$$d \text{tr } W^j = jt \sum_{h=1}^{e_n} \lambda_h^{j-1} d\lambda_h.$$

The first  $e_n$  rows give an invertible Vandermonde matrix. Hence the trace Jacobian and the eigenvalue Jacobian have the same rank there, including over a commuting Taylor fraction field. We have therefore obtained both ingredients needed for the response calculations: scalar branches and specialization-stable rank bounds.

## 5 The space of eigenvalue responses

Fix a size  $n \in \mathcal{S}$ , a generic input pair  $P$ , and a splitting field  $\Omega_P$  of  $W(P)$  over  $\text{Frac } K[[P]]$ . Write its distinct roots as  $\lambda_h$ , with spectral projections  $\pi_h$ ,  $1 \leq h \leq e_n$ . Each projection has rank  $t$ . We also use these data after embedding into the valued fields of Section 4.

We seek the possible leading changes of the eigenvalues at  $P$ . Besides varying  $P$  itself, we allow an excursion through another input block and a return to  $P$ . Their linear span will determine which eigenvalues can be varied independently.

### 5.1 Loop and paired responses

Upper triangular block evaluations are also the basic mechanism of noncommutative difference-differential calculus; see [9, Proposition 2.2 and Theorem 3.11]. Here the admissible evaluations of Section 4 let us extract scalar eigenvalue responses from those block calculations.

For a perturbation  $\mathcal{W}$  of  $W(P)$ , put  $\pi = \pi_h$ ,  $\bar{\pi} = 1 - \pi$ , and

$$C_h(\mathcal{W}) = \bar{\pi}(\mathcal{W} - \lambda_h)\bar{\pi}.$$

This operator acts on the complementary space. Its constant term is invertible, so its inverse has a unique expansion in either commuting or free formal parameters. Define the fixed-root Schur expression by

$$S_h(\mathcal{W}) = \pi(\mathcal{W} - \lambda_h)\pi - \pi\mathcal{W}\bar{\pi}C_h(\mathcal{W})^{-1}\bar{\pi}\mathcal{W}\pi. \quad (5.1)$$

This expression records the obstruction to keeping the root  $\lambda_h$  fixed. For  $u$  in the image of  $\pi$ , the vector

$$u - C_h(\mathcal{W})^{-1}\bar{\pi}\mathcal{W}\pi u$$

is the unique vector projecting to  $u$  whose complementary component under  $\mathcal{W} - \lambda_h$  vanishes. Its remaining component is  $S_h(\mathcal{W})u$ . Thus  $S_h(\mathcal{W}) = 0$  means that the fixed-root kernel is a graph over the original eigenspace.

When the constant output is split semisimple and these equations are imposed for every one of its distinct roots, the kernel graphs combine into a formal change of basis. The output remains conjugate to its constant term and satisfies that constant term's minimal polynomial. Lemma 10.2 proves the equivalence with the polynomial identity at every finite word order.

The same Schur expression is defined near  $\text{diag}(P, C_0)$  if  $\lambda_h$  is absent from  $W(C_0)$ , embedding  $\pi_h$  at the  $P$  block. In this use, equations indexed by labels at  $P$  record those fixed roots only.

For a loop direction  $Y = (Y_1, Y_2)$  at  $P$ , define the scalar  $D_h(P; Y)$  by

$$\pi_h W'_P(Y) \pi_h = D_h(P; Y) \pi_h. \quad (5.2)$$

This compression is scalar: apply Lemma 4.6 to the commuting deformation  $P + xY$  and take its first coefficient. Equivalently,  $D_h(P; Y)$  is the differential of the branch  $\lambda_h$  in direction  $Y$ . By Lemma 4.4, each of these linear functionals is nonzero. Because  $K$  is infinite, we can choose a constant direction  $v$  over  $K$  with

$$D_h(P; v) \neq 0 \quad (1 \leq h \leq e_n). \quad (5.3)$$

Indeed the forbidden directions form a finite union of proper  $K$ -linear subspaces, even though their defining coefficients belong to  $\Omega_P$ .

Let  $Q$  be an independent generic point of size  $n' \in \mathcal{S}$ . Let  $U$  be an input arrow  $P \leftarrow Q$  and  $V$  an input arrow  $Q \leftarrow P$ ; each is a pair of rectangular scalar matrices. Form the triangular input pair

$$T = \begin{pmatrix} P & U & 0 \\ 0 & Q & V \\ 0 & 0 & P \end{pmatrix}.$$

Set

$$M_h(P, Q; U, V) = \pi_h \left( W(T)_{13} - W(T)_{12}(W(Q) - \lambda_h)^{-1}W(T)_{23} \right) \pi_h. \quad (5.4)$$

Block paths show that this is bilinear in  $U, V$ . The inverse exists because the spectra at independent points are disjoint.

**Lemma 5.1.** *There is a scalar  $m_h(P, Q; U, V)$  such that*

$$M_h(P, Q; U, V) = m_h(P, Q; U, V) \pi_h.$$

*It is the coefficient of  $xy$  in the branch reducing to  $\lambda_h$  for the commuting input deformation*

$$\begin{pmatrix} P & xU \\ yV & Q \end{pmatrix}.$$

*The same assertion holds at an admissible specialized pair whose spectra are disjoint and whose combined output is split semisimple with the maximal number  $e_{n+n'}$  of distinct roots.*

*Proof.* At  $x = y = 0$ , disjointness and Lemma 4.5 give  $e_n + e_{n'} = e_{n+n'}$  roots. Thus Lemma 4.6 applies. Either arrow alone leaves the spectrum unchanged, so the chosen branch has no terms depending on  $x$  alone or on  $y$  alone.

Compute the  $xy$  coefficient of its fixed-root Schur expression. The direct return from  $P$  through  $Q$  gives  $W(T)_{13}$ . Eliminating the  $Q$  block gives the second term in (5.4). The other eigenspaces in the  $P$  block give no further term of this order: their couplings to  $\pi_h$  have zero constant term and no first-order term under these off-diagonal input perturbations. Replacing the fixed root by the scalar branch subtracts its  $xy$  coefficient times  $\pi_h$ . The resulting Schur expression vanishes, because the branch eigenspace projects isomorphically to the base eigenspace. This proves the formula. The specialized assertion uses exactly the same argument and the specialization-stable degree bound.  $\square$

## 5.2 Coefficient spans and their specializations

At  $Q = 0$  the matrices  $W(Q) - \lambda_h$  are invertible, since no  $\lambda_h$  belongs to the spectrum of  $w(0)$ . Hence (5.4) has a formal expansion in the entries of  $Q$  over  $\Omega_P$ . Define

$$m_h = \frac{1}{t} \operatorname{tr} M_h, \quad m = (m_h)_{h=1}^{e_n}, \quad D = (D_h)_{h=1}^{e_n}.$$

Here  $D = D(P; \cdot) : M_n(\Omega_P)^2 \rightarrow \Omega_P^{e_n}$  is the linear map of loop responses. Its columns are its values on the standard coordinate directions in the two input matrices. The scalar identity of Lemma 5.1 holds also for these formal expansions, by clearing their determinants and applying Lemma 4.2.

Define  $\mathcal{T}_P \subseteq \Omega_P^{e_n}$  to be the linear span of the columns of  $D$  and of every coefficient vector of  $m$ , for every allowed partner size  $n' \in \mathcal{S}$  and every pair of coordinate arrows  $U, V$ . Write

$$a_n = \dim_{\Omega_P} \mathcal{T}_P.$$

Although the definition permits infinitely many partner sizes and coefficients, the ambient space is finite dimensional.



**Lemma 5.2** (Specialization and finite witnesses). *The scalar identities for  $M_h$  and every linear identity expressing  $m \in \mathcal{T}_P$  hold after any admissible specialization of the partner input for which the required inverses exist. For identities on a subset of labels, only the inverses for those labels are needed.*

*After a field extension, the columns of  $D$  and finitely many paired responses span  $\mathcal{T}_P$ . These responses can be taken at independent small generic partner points, whose sizes need not be equal.*

*Proof.* For the first assertion, apply any  $\Omega_P$ -linear functional annihilating  $\mathcal{T}_P$  to the formal tuple  $m$ . The result is zero. Multiply by the finitely many determinants of  $W(Q) - \lambda_h$  that occur. The cleared expression is a finite sum of products of base-point scalars and matrix-entry series; its auxiliary coefficients have a uniform valuation bound. Its formal identity is therefore equivalent to its identity at a small generic partner, by Lemma 4.2. It persists under the admissible specializations of Lemma 4.1. Divide by the specialized determinants when they are nonzero. The same argument, entry by entry, proves the scalar identities for  $M_h$ . It also proves the projected assertion, without introducing any spectral labels for the partner input.

Choose finitely many coefficient vectors, together with columns of  $D$ , that span  $\mathcal{T}_P$ , and select a nonzero minor witnessing their rank. For each column requiring a partner, use a separate set of partner variables. By multilinearity, the corresponding determinant of the series columns has that nonzero minor as the coefficient of a specified product of monomials. It is therefore a nonzero formal series. Clear the column denominators. The numerator is still nonzero, and its coefficients are uniformly bounded below in valuation. Independent small generic evaluations keep this numerator and its denominators nonzero, by Lemma 4.2. The resulting values attain the full dimension. The first assertion places all of them in the extended active space.

This argument evaluates cleared numerators, not the possibly divergent geometric expansions of their inverses. The distinction is precisely the one made in Section 4.1.  $\square$

The lemma also applies after adjoining Taylor or Laurent parameters. One first clears determinants and then works coefficientwise as in Lemma 4.1. In particular, it permits triangular partners with repeated base blocks, provided the inverses required for the selected target roots exist in the chosen Laurent field.

### 5.3 A partition of the eigenvalue labels

The active space has a stronger structure than a general linear subspace. Its coordinates form groups whose responses are always in a fixed ratio.

**Proposition 5.3.** *For the direction  $v$  chosen in (5.3), there is a partition of  $\{1, \dots, e_n\}$  such that*

$$\mathcal{T}_P = \left\{ c \in \Omega_P^{e_n} : \frac{c_h}{D_h(P; v)} \text{ is constant on each part} \right\}.$$

*Proof.* We first prove that  $\mathcal{T}_P$  is closed under the operation

$$(a, b) \mapsto \left( \frac{a_h b_h}{D_h(P; v)} \right)_h. \quad (5.5)$$

Choose two finite spanning sets supplied by Lemma 5.2, using independent partner points for the two sets. It suffices to prove the assertion for a column  $a$  in the first set and a column  $b$  in the second, over their joint coefficient field  $E$ .

Put  $P' = P + \xi v$ . Use a triangular chain with endpoints  $P, P$ , middle point  $P'$ , an optional point  $Q_1$  between the first endpoint and  $P'$ , and an optional point  $Q_2$  between  $P'$  and the last



endpoint. A direct step represents a loop column; an excursion through  $Q_i$  represents a paired column. Put the arrows defining  $a$  and  $b$  on the successive steps, with no other input arrows. Group the internal positions as a single partner  $C(\xi)$ . Its size belongs to  $\mathcal{S}$ .

For every target label,  $W(C(\xi)) - \lambda_h$  is invertible over  $E((\xi))$ . The optional blocks have spectra disjoint from  $P$ . At the middle block, the branch reducing to  $\lambda_h$  moves with nonzero derivative  $D_h(P; v)$ ; all other branches are already away from  $\lambda_h$ . Thus

$$\xi(W(P') - \lambda_h)^{-1} \Big|_{\xi=0} = \frac{\pi_h}{D_h(P; v)}. \quad (5.6)$$

This follows directly by scalar branch separation.

By Lemma 5.2, the paired response through  $C(\xi)$  belongs to  $\mathcal{T}_P \otimes E((\xi))$ . Eliminate the optional blocks first in its Schur formula. Triangularity means that their elimination does not change the middle diagonal block. All terms are regular at  $\xi = 0$  except the middle inverse in (5.6). The effective first-to-middle coupling, compressed by  $\pi_h$  at both ends and evaluated at zero, is  $a_h \pi_h$ . For a direct step this is (5.2); for an excursion it is (5.4). The corresponding middle-to-last coupling is  $b_h \pi_h$ .

The singular term occurs with the minus sign in the Schur complement. Multiplying by  $\xi$  and reducing at zero therefore gives

$$\left( -\frac{a_h b_h}{D_h(P; v)} \right)_h \in \mathcal{T}_P \otimes E.$$

Taking a Laurent coefficient preserves membership in this fixed finite dimensional space: apply its annihilating linear functionals. Bilinearity and the two spanning sets prove closure under (5.5) after extension to  $E$ , and faithful scalar extension proves it over  $\Omega_P$ .

Divide coordinates by the nowhere-zero vector  $D(P; v)$ . The resulting space  $\mathcal{U} \subseteq \Omega_P^{e_n}$  contains 1, since  $D(P; v) \in \mathcal{T}_P$ , and is closed under coordinatewise multiplication. Declare two coordinates equivalent if every element of  $\mathcal{U}$  has the same value on them. By avoiding finitely many proper linear subspaces, choose an element of  $\mathcal{U}$  whose values distinguish all the equivalence classes. Lagrange interpolation in this element gives the indicator function of each class. These indicators span all class-constant functions. Thus  $\mathcal{U}$  is exactly that algebra of functions, proving the proposition.  $\square$

We call two labels in the same part *linked*. The number of parts is  $a_n$ . Selecting one representative from each part identifies  $\mathcal{T}_P$  with the full coordinate space on those representatives. The next stage must show that these response relations control higher-order free-letter errors, rather than only the two elementary perturbations used to define them.

## 6 Transfer of the first unsolved word coefficient

Our next objective is to control equations in free letters. The first and paired responses describe matrix perturbations with one loop or two opposite arrows. We show that they also contain the first unsolved coefficient of an arbitrary free-series perturbation. The argument unfolds a word along a triangular chain and then replaces its internal states by a single auxiliary partner. The only difficulty is that this partner initially has the target eigenvalue. The following linear-algebra observation explains how to remove that singularity without changing the coefficient being computed.

**Lemma 6.1** (An inverse between two vanishing couplings). *Let  $E$  be a field and let  $M(\xi) \in \text{End}_E(V)[[\xi]]$ , where  $V$  is finite dimensional. Suppose that zero is a semisimple eigenvalue of  $M_0 = M(0)$ , and let  $\Pi$  be its spectral projection. Suppose that*

$$G = \Pi M'(0) \Pi|_{\ker M_0}$$

*is invertible. Then  $M(\xi)^{-1}$  exists over  $E((\xi))$  and has a pole of order at most one. Let  $J$  be zero on  $\ker M_0$  and equal to  $M_0^{-1}$  on  $\text{im } M_0$ . If  $L(\xi)$  and  $B(\xi)$  are compatible matrix series satisfying*

$$L(0)\Pi = 0, \quad \Pi B(0) = 0,$$

*then*

$$\lim_{\xi \rightarrow 0} L(\xi) M(\xi)^{-1} B(\xi) = L(0) J B(0). \quad (6.1)$$

*Suppose additionally that  $e$  is an idempotent,  $f = 1 - e$ ,  $f M_0 f$  is invertible on  $fV$ , and  $e : \ker M_0 \rightarrow eV$  is an isomorphism. Under the same two coupling conditions,*

$$L(0) J B(0) = L(0) f (f M_0 f)^{-1} f B(0). \quad (6.2)$$

*Proof.* Use  $V = \ker M_0 \oplus \text{im } M_0$ . In this decomposition

$$M(\xi) = \begin{pmatrix} \xi G + O(\xi^2) & O(\xi) \\ O(\xi) & M_0|_{\text{im } M_0} + O(\xi) \end{pmatrix}.$$

The lower-right block is invertible and the Schur complement of that block is  $\xi G + O(\xi^2)$ . The upper-left block of  $M(\xi)^{-1}$  is consequently  $\xi^{-1} G^{-1} + O(1)$ ; its off-diagonal blocks are bounded; and its lower-right block tends to  $M_0^{-1}|_{\text{im } M_0}$ . The kernel components of both couplings are  $O(\xi)$ . Multiplying these four blocks proves (6.1).

For the second assertion, write  $b = B(0)u$ . The condition  $\Pi b = 0$  places  $b$  in  $\text{im } M_0$ . Solutions of  $M_0 x = b$  differ by  $\ker M_0$ . Since projection to  $eV$  is an isomorphism on that kernel, there is a unique solution with  $ex = 0$ . Its  $f$  coordinate is  $(f M_0 f)^{-1} f b$ . All solutions have the same image under  $L(0)$  because  $L(0)$  kills  $\ker M_0$ . Comparing this solution with  $Jb$  proves (6.2).  $\square$

We now recall exactly which response identities enter the transfer argument. At a small generic point  $P$  of size  $n \in \mathcal{S}$ , the eigenvalues  $\lambda_h$  have projections  $\pi_h$  of rank  $t$ . The first response is  $D_h(P; Y) \pi_h$ . A pair of arrows through an independent partner  $Q$  gives  $m_h(P, Q; U, V) \pi_h$ . The columns of the first responses and the coefficient vectors of all these paired responses span  $\mathcal{T}_P$ . By Lemma 5.2, their scalarity and all linear containment identities persist for a block-triangular partner after its finitely many resolvent determinants are cleared. For a set of target labels  $I$ , only the determinants belonging to labels in  $I$  are needed. We use the notation

$$\mathcal{T}_P|_I = \{(c_h)_{h \in I} : (c_h)_h \in \mathcal{T}_P\}.$$

These spaces and identities are extended to the coefficient field in use. We also fix a direction  $v$  with  $D_h(P; v) \neq 0$  for every target label.

**Proposition 6.2** (Transfer of a leading coefficient). *Let  $R = \text{diag}(P, C_0)$  be a small matrix input of total size  $N \in \mathcal{S}$ ; the block  $C_0$  may be absent. Let  $I$  be a set of labels at  $P$  such that  $W(C_0) - \lambda_h$  is invertible for every  $h \in I$ . Work over a field  $E$  containing the labeled spectral data at  $P$  and the entries of  $C_0$ , with the formal evaluations allowed by Lemma 4.1. Let*

$$Z(s_1, s_2) \in M_N(E) \langle\langle s_1, s_2 \rangle\rangle^2$$

have constant term  $R$ . These are input matrices: their scalar entries act on the input factor and commute with the old coefficient algebra.

For  $h \in I$ , embed  $\pi_h$  at the  $P$  block, put  $\bar{\pi}_h = 1 - \pi_h$ , and form

$$S_h(Z) = \pi_h(W(Z) - \lambda_h)\pi_h - \pi_h W(Z) \bar{\pi}_h (\bar{\pi}_h(W(Z) - \lambda_h) \bar{\pi}_h)^{-1} \bar{\pi}_h W(Z) \pi_h.$$

Suppose that every  $S_h(Z)$  vanishes in degrees less than  $\ell$ , where  $\ell \geq 1$ . For each word  $\omega$  of length  $\ell$ , there are scalars  $c_{h,\omega} \in E$  such that

$$[S_h(Z)]_\omega = c_{h,\omega} \pi_h, \quad (c_{h,\omega})_{h \in I} \in (\mathcal{T}_P|_I) \otimes E.$$

The same conclusion holds for homogeneous degree- $\ell$  coefficients of perturbations in finitely many ordinary commuting parameters.

*Proof.* For  $\ell = 1$ , only the  $P$ -to- $P$  input coefficient contributes to the compressed derivative, so the assertion is the first-response identity. Assume  $\ell \geq 2$  and fix a word  $\omega$  of length  $\ell$ .

**Unfolding the word.** Use  $\ell + 1$  ordered states. On consecutive steps put the letters of  $\omega$ , and evaluate a word by following that path. Thus a coefficient of  $Z$  for a substring becomes the corresponding strictly upper block of the input, while every diagonal block is  $R$ . The  $(1, \ell + 1)$  block extracts precisely the coefficient of  $\omega$ . This substitution is a homomorphism on the free-series ring modulo degree  $\ell + 1$ , and it therefore respects every inverse with invertible constant term.

Keep only  $P$  at the two endpoint states, leaving all  $R$  blocks at the  $\ell - 1$  internal states. This operation does not change the required compressed endpoint coefficient. At an endpoint, constant input steps preserve the blocks  $P, C_0$ , and the constant output commutes with  $\pi_h$ . A contributing Schur path can therefore neither enter the first endpoint complement from its target projection nor leave the last endpoint complement for its target projection. All complement eliminations relevant to the coefficient occur at internal states.

Denote the internal input by  $Q_0$ . Its size is  $(\ell - 1)N$  and hence belongs to  $\mathcal{S}$ . Grouping the internal states, write the cropped input as

$$T_0 = \begin{pmatrix} P & G_1 & Y \\ 0 & Q_0 & G_2 \\ 0 & 0 & P \end{pmatrix}.$$

Here each entry denotes a pair of input matrices. In the output write  $E_1 = W(T_0)_{12}$ ,  $E_2 = W(T_0)_{23}$ , and  $F = W(T_0)_{13}$ .

**The internal target eigenspace.** Fix  $h \in I$ , set  $M_0 = W(Q_0) - \lambda_h$ , and let  $e$  be the sum of the fixed copies of  $\pi_h$  on its internal states. Put  $f = 1 - e$ . The matrix  $fM_0f$  is upper triangular with invertible diagonal blocks. Moreover its Schur complement on  $eV$  is zero: every strictly upper coefficient is a coefficient for a proper substring of  $\omega$ , and these vanish by hypothesis. Its kernel is consequently the graph

$$u \longmapsto u - f(fM_0f)^{-1}fM_0eu, \quad u \in eV.$$

Projection onto  $eV$  is an isomorphism on this kernel, whose dimension is  $(\ell - 1)t$ . Triangularity shows that this is also the full algebraic multiplicity of zero. Thus zero is semisimple. Let  $\Pi$  be its spectral projection.

The same argument applies to the chain consisting of the first endpoint and the internal states, and to the chain consisting of the internal states and the last endpoint. They involve only proper substrings. Solving their triangular kernel equations shows that the couplings between their zero eigenspaces vanish:

$$\pi_h E_1 \Pi = 0, \quad \Pi E_2 \pi_h = 0. \quad (6.3)$$

Indeed any nonzero compressed coupling would reduce the kernel dimension below the sum of the diagonal zero multiplicities. This argument uses semisimplicity only at the target eigenvalue.

**A regularized auxiliary partner.** In every internal copy of  $R$ , replace its  $P$  block by  $P + \xi v$ , keeping all arrows fixed; call the resulting input  $Q(\xi)$ . Put  $M(\xi) = W(Q(\xi)) - \lambda_h$ . The spectral projection  $\Pi$  is upper triangular in the state order, with diagonal equal to the fixed target projections. The graph identification of its range with  $eV$  is upper triangular with identity diagonal. It follows that

$$\Pi M'(0) \Pi|_{\ker M_0}$$

is upper triangular in this identification, with every diagonal block equal to  $D_h(P; v) I_t$ . It is therefore invertible. This verifies the simple-pole hypothesis of Lemma 6.1, including the effect of all internal triangular arrows.

Let  $T(\xi)$  be the corresponding full cropped input and use  $E_1(\xi)$ ,  $E_2(\xi)$ ,  $F(\xi)$  for its output blocks. For nonzero formal  $\xi$ , the compressed endpoint expression is

$$\begin{aligned} \mathcal{E}_h(\xi) &= \pi_h (F(\xi) - E_1(\xi) M(\xi)^{-1} E_2(\xi)) \pi_h \\ &= D_h(P; Y) \pi_h + M_h(P, Q(\xi); G_1, G_2). \end{aligned} \quad (6.4)$$

Its tuple is scalar and belongs to the projected active space over  $E((\xi))$ . This is exactly the specialized paired-response identity: the auxiliary block is upper triangular, its diagonal entries are small blocks with a Taylor shift, and its strictly upper entries are fixed finite matrices. Clear the target resolvent determinants first and apply Lemma 5.2; restore the inverses over  $E((\xi))$ , where the preceding argument proves they exist. No specialization of an undefined inverse is being used.

**Identifying the limit.** By (6.3) and Lemma 6.1, expression (6.4) has the finite limit

$$\pi_h (F - E_1 J E_2) \pi_h = \pi_h (F - E_1 f (f M_0 f)^{-1} f E_2) \pi_h.$$

The last expression eliminates precisely the fixed complementary spaces in the internal states. By the unfolding and endpoint-cropping argument it is  $[S_h(Z)]_\omega$ . In particular the limiting inverse has not changed the coefficient to one computed with a different choice of complement. Taking the limit in the scalar-response identities and in the finitely many linear equations defining  $\mathcal{T}_P|_I$  proves the assertion simultaneously for all target labels.

Finally consider a perturbation in commuting variables  $h_1, \dots, h_s$ . Substitute  $h_j = a_j z$  for arbitrary scalar directions  $a_j$  and one free letter  $z$ . The lower homogeneous terms still vanish, so the result just proved applies to the degree- $\ell$  coefficient on every such line. Scalarity and containment in a fixed finite-dimensional linear space are linear conditions. Each of their annihilators therefore gives a homogeneous polynomial in the  $a_j$  that vanishes identically. Over the infinite characteristic-zero field all its coefficients vanish. This proves the commuting-parameter assertion.  $\square$

Proposition 6.2 is the bridge from responses to free-series equations. It controls a first unsolved coefficient without assuming that the perturbation is commutative or that the auxiliary matrix has a separated spectrum before regularization. We next use its commuting-parameter consequence to show that linked responses arise from actual formal relations between eigenvalues.

## 7 Functional linkage and replication

We now show that the dimension of the active response space is additive under generic direct sums. The main point is to turn the linear relations between linked responses into identities between eigenvalue branches. These identities will continue to control responses through partners of every allowed size.

Retain the progression  $\mathcal{S} = \{pN : N \geq 1\}$  and the common eigenvalue multiplicity  $t$ . Thus, at size  $n \in \mathcal{S}$ , the number of distinct generic eigenvalues is  $e_n = dn/t$ . We use the following previously established facts. At a direct sum of independent small generic inputs of sizes in  $\mathcal{S}$ , the spectra are disjoint and have the maximal number of distinct eigenvalues. Their commuting Taylor deformations therefore have scalar spectral blocks, by Lemma 4.6. The response identities admit the specializations of Lemma 5.2, and their leading terms satisfy Proposition 6.2.

For a generic input  $P$ , let  $\Omega_P$  be its spectral splitting field, and choose a constant input direction  $v$  such that  $D_h(P; v) \neq 0$  for every eigenvalue label  $h$ . Recall the active-space partition  $\mathcal{P}_P$ , characterized by

$$\mathcal{T}_P = \left\{ c \in \Omega_P^{e_n} : \frac{c_h}{D_h(P; v)} \text{ is constant on each member of } \mathcal{P}_P \right\}. \quad (7.1)$$

Labels in one member of this partition are called linked. In particular,  $a_n = \dim_{\Omega_P} \mathcal{T}_P$  is the number of its members. The argument below uses the full coefficientwise definition of  $\mathcal{T}_P$ : it contains the loop responses  $D(P; Y)$  and the coefficient vectors of  $m(P, Q; U, V)$  for every partner size in  $\mathcal{S}$ .

### 7.1 A formal identity for linked branches

**Proposition 7.1** (Functional linkage). *Let  $P$  be a small generic input of size  $p$ , and let  $h, h'$  be linked labels at  $P$ . Let  $C_1, \dots, C_s$  be small generic inputs of sizes in  $\mathcal{S}$ , jointly independent with  $P$ , where  $s = 0$  is allowed. Put  $R = \text{diag}(P, C_1, \dots, C_s)$ . Over a common spectral splitting field  $E$ , denote by  $\Lambda_j^R(\Delta)$  the scalar eigenvalue branch of  $W(R + \Delta)$  whose value at zero is  $\lambda_j(P)$ , for  $j = h, h'$ .*

*Define*

$$g_j(\eta) = \lambda_j(P + \eta v) - \lambda_j(P) \in \Omega_P[[\eta]], \quad j = h, h'.$$

*Then  $g_{h'}'(0) \neq 0$ , and*

$$\Lambda_h^R(\Delta) - \lambda_h(P) = (g_h \circ g_{h'}^{-1})(\Lambda_{h'}^R(\Delta) - \lambda_{h'}(P)). \quad (7.2)$$

*Here  $g_{h'}^{-1}$  denotes the inverse for composition. In particular, the univariate series in this identity depends only on  $P, h, h'$  and  $v$ , and is unchanged when auxiliary blocks are adjoined.*

*Proof.* We first explain why the response relations persist while  $P$  moves. By the Taylor convention in Section 4.1, substitution  $P \mapsto P + \eta v$  gives homomorphisms

$$\text{Frac } K[[P]] \longrightarrow (\text{Frac } K[[P]])[[\eta]], \quad \Omega_P \longrightarrow \Omega_P[[\eta]],$$

both reducing to the identity at  $\eta = 0$ . The second uniquely extends the first through the finite separable extension  $\Omega_P / \text{Frac } K[[P]]$ . Thus every nonzero element has image with invertible constant term, and the eigenvalues and projections map to their uniquely labeled Taylor deformations.

For linked labels, Equation (7.1) gives

$$D_{h'}(P; v)c_h - D_h(P; v)c_{h'} = 0 \quad (7.3)$$

for every loop response and every coefficient vector of every paired response. Apply the preceding homomorphism coefficientwise in the partner coordinates. The resulting identities are exactly Equation (7.3) at the moved input  $P + \eta v$ , with the moved eigenvalues, projections, and  $D$ -factors. They remain valid in the triangular specializations used in Proposition 6.2. More explicitly, first clear the finitely many auxiliary resolvent determinants. At each Taylor degree, the number of strictly upper steps is bounded. This permits their entries to have Laurent coefficients in  $\eta$ , while the diagonal entries are expanded at the original small inputs. Coefficientwise convergence and specialization are then supplied by Lemma 5.2. Thus the transfer argument applies over  $E((\eta))$  with the transported response relation.

Write  $v_R$  for  $v$  supported in the  $P$ -block and put  $R(\eta) = R + \eta v_R$ . At  $R(\eta) + H$ , impose the equation

$$\Lambda_{h'}^{R(\eta)}(H) = \lambda_{h'}(P + \eta v). \quad (7.4)$$

Choose linear coordinates  $H = uv_R + H_{\text{rest}}$ , with  $H_{\text{rest}}$  in a fixed complementary space of input directions. The derivative of the left side in  $u$  at  $H = 0$  is  $D_{h'}(P + \eta v; v)$ , a unit in  $E[[\eta]]$ . Formal implicit substitution therefore eliminates  $u$ . In particular, the quotient of  $E[[\eta, H]]$  by Equation (7.4) is

$$E[[\eta, H_{\text{rest}}]], \quad E[[\eta, H_{\text{rest}}]] \hookrightarrow E((\eta))[[H_{\text{rest}}]]. \quad (7.5)$$

The eliminated series has zero constant term in  $H_{\text{rest}}$ .

On this quotient, scalar-block separation implies that the fixed-root Schur expression  $S_{h'}$  at  $R(\eta)$  vanishes. We claim that  $S_h$  also vanishes over the right-hand ring in Equation (7.5). Induct on total degree in  $H_{\text{rest}}$ . Both Schur expressions vanish in degree zero. If they vanish below degree  $l$ , the commuting-parameter conclusion of Proposition 6.2, together with the transported relation (7.3), gives scalar leading coefficients  $[S_j]_l = c_{j,l}\pi_j$  for  $j = h, h'$ , with

$$D_{h'}(P + \eta v; v)c_{h,l} = D_h(P + \eta v; v)c_{h',l} = 0.$$

Here the  $c_{j,l}$  are scalar homogeneous polynomials in the remaining parameters over  $E((\eta))$ ; the projections are the moved target projections. The first factor is nonzero, so  $c_{h,l} = 0$  and  $[S_h]_l = 0$ .

It follows that the  $h$ -branch is fixed as well:

$$\Lambda_h^{R(\eta)}(H) = \lambda_h(P + \eta v)$$

on the quotient. For clarity, the implication from  $S_h = 0$  uses the scalar blocks of Lemma 4.6: the complementary block at the fixed root is invertible, and the determinant of the full matrix at that root is a unit times the  $t$ -th power of the shift of the  $h$ -branch. The quotient is a domain, so the shift is zero. Both branch series and the eliminated coordinate are regular in  $\eta$ . The injection in Equation (7.5) therefore descends this equality to  $E[[\eta, H_{\text{rest}}]]$ .

Finally make the invertible linear coordinate change  $\Delta = H + \eta v_R$ . Equation (7.4) becomes

$$\Lambda_{h'}^R(\Delta) - \lambda_{h'}(P) = g_{h'}(\eta).$$

Since  $g_{h'}'(0) = D_{h'}(P; v) \neq 0$ , it eliminates  $\eta$  as  $g_{h'}^{-1}(\Lambda_{h'}^R(\Delta) - \lambda_{h'}(P))$ . Substitution in the equality for the  $h$ -branch gives Equation (7.2).  $\square$

## 7.2 All partner sizes and the dimension formula

We next pass from an identity near a direct sum to an identity for every coefficient defining the generic active space. This step is necessary because a bound proved using a fixed collection of partners would not bound  $a_n$ .

**Proposition 7.2** (Replication of active dimensions). *For every positive integer  $N$ ,*

$$a_{pN} = Na_p.$$

*Proof.* Put  $n = pN$  and choose independent small generic inputs  $P_1, \dots, P_N$  of size  $p$ . Write  $R_1 = \text{diag}(P_1, \dots, P_N)$  and use  $x = R_1 + \Delta_1$  to label the generic size- $n$  spectrum by its branches at  $R_1$ .

This substitution embeds  $K[[x]]$  in  $E[[\Delta_1]]$ . Indeed, for an image of a series over  $K$ , the Taylor coefficients are uniformly integral in the small parameter  $\varepsilon$ . Substituting  $\Delta_1 = \varepsilon X - R_1$ , with independent coordinates  $X$ , is therefore convergent and recovers the original series at  $\varepsilon X$ . Homogeneous expansion at that point detects every nonzero series. The embedding extends to the fraction field. All generic spectral roots are supplied by the scalar branches at  $R_1$ , so it also embeds a spectral splitting field  $\Omega_x$  into  $\text{Frac } E[[\Delta_1]]$ , with these labels. The branches and spectral projections themselves are regular at  $\Delta_1 = 0$ .

Fix a block  $i$  and labels  $h, h'$  in one member of  $\mathcal{P}_{P_i}$ . Choose a constant direction  $v_i$  supported in that block, nonzero under every  $D_h(P_i; \cdot)$ . Apply Proposition 7.1 first at  $R_1$ . Differentiating its identity gives proportional loop responses at  $x$ . Now adjoin an independent generic partner  $Q$  of any fixed size in  $\mathcal{S}$ , and apply the same proposition to  $\text{diag}(R_1, Q)$ . Its univariate series is the same one. At  $\text{diag}(x, Q)$ , perturb the opposite input blocks by  $uU$  and  $zV$ . Either arrow alone preserves the spectrum, so the first derivatives in  $u$  and  $z$  vanish. The coefficient of  $uz$  in the branch is the paired response  $m_h(x, Q; U, V)$ . The mixed chain rule and the separate first derivative in  $v_i$  give

$$D_{h'}(x; v_i) m_h(x, Q; U, V) - D_h(x; v_i) m_{h'}(x, Q; U, V) = 0. \quad (7.6)$$

The two  $D$ -factors have nonzero reductions at  $\Delta_1 = 0$ .

We must verify this relation coefficientwise in  $Q$ , rather than only at its generic value. Cross multiplication has already removed the response ratio. Multiply further by the finitely many determinants of  $W(Q) - \lambda_h(x)$  and  $W(Q) - \lambda_{h'}(x)$  needed to clear the auxiliary inverses. At each fixed  $\Delta_1$ -degree, the resulting series in the independent coordinates of  $Q$  has coefficients uniformly bounded below in  $\varepsilon$ -valuation. The branch and projection coefficients do not involve  $Q$ ; the other factors are ordinary matrix series with two prescribed arrow steps and a fixed number of Taylor insertions. Their path expansions give precisely this bound, as in Lemma 4.1. Evaluation at small generic  $Q$  therefore detects every coefficient. Equation (7.6) is consequently a formal identity in those coordinates. Injectivity of the embedding of  $\Omega_x$  descends it to the generic size- $n$  field. Finally, the cleared determinants have nonzero constant terms in  $Q$ , so they can be canceled in  $\Omega_x[[Q]]$ .

This argument applies separately to every partner size with the same labeling of  $x$  and the same  $D$ -factors. It therefore gives a relation on every coefficient vector used to define  $\mathcal{T}_x$ . Together with the corresponding loop relations, it says that  $c_h/D_h(x; v_i)$  is constant on each old partition member within block  $i$ , for every  $c \in \mathcal{T}_x$ . These partitions have  $Na_p$  members in total, and all their normalizing factors are nonzero. Thus  $a_n \leq Na_p$ .

For the opposite inequality, use Lemma 5.2 for each  $P_i$ , choosing all partners jointly independent. At  $R_1$ , support each loop direction, and both arrows of each paired response, in its chosen block  $i$ . The responses then coincide with the original responses on that block and vanish on every other block. They have rank  $Na_p$ . Evaluate the same finite collection first at  $R_1 + \Delta_1$ . All branch data and partner resolvents are regular at  $\Delta_1 = 0$ , since the partner spectra are disjoint from that of  $R_1$ . A generic rank bound smaller than  $Na_p$  would make every minor of that order vanish over the embedded generic field, and hence also at the reduction. The displayed specialization contradicts this. Therefore  $a_n \geq Na_p$ .  $\square$



### 7.3 Partners of one common size

The dimension formula permits us to use a fixed matrix size on both ends of every paired response. We will also need to fix one arrow and vary only its reverse.

**Corollary 7.3** (Equal-size partners). *There are  $r \in \mathcal{S}$  and a positive integer  $K_0$  with the following property. Let  $P, Q_1, \dots, Q_{K_0}$  be independent small generic inputs of size  $r$ . For generic fixed input arrows  $U_j : P \leftarrow Q_j$ , the loop columns and the images of*

$$V \mapsto m(P, Q_j; U_j, V), \quad V : Q_j \leftarrow P, \quad 1 \leq j \leq K_0,$$

*span  $\mathcal{T}_P$  after extension to the joint field of the data. The analogous assertion holds with the other arrow fixed. The same  $r$  and  $K_0$  may be used for both assertions.*

*Proof.* Choose a finite witness list at size  $p$  from Lemma 5.2, and let its partner sizes be  $q_1, \dots, q_M \in \mathcal{S}$ . Choose  $r = pN$  with  $r \geq q_j$  for every  $j$ . Specialize the size- $r$  base to a direct sum of  $N$  independent size- $p$  inputs. For each block, use a fresh copy of the witness list. Pad a size- $q_j$  partner to size  $r$  by adjoining independent generic diagonal data of size  $r - q_j$ , omitting the padding if this difference is zero. Give both arrows zero entries on the padding. The response remains the original size- $p$  witness on its chosen base block and is zero on the other base blocks.

The resulting columns have rank  $Na_p = a_r$  by Proposition 7.2. All inputs at this specialization have maximal spectral count, and the partner spectra are disjoint from the base spectrum. The regular-minor argument in the preceding proof therefore transfers this rank to generic size- $r$  base and partner inputs.

Assign a separate partner to every paired column in this finite list. At the witness specialization, choose its forward arrow to be the specified  $U_j$  and test the variable reverse arrow on the specified  $V_j$ . A minor of order  $a_r$  is nonzero there. Since paired responses are bilinear in their arrows, this minor remains nonzero for generic fixed forward arrows and suitable reverse-arrow columns. This proves the first assertion with finitely many partners. The identical argument fixes the reverse arrows instead. Increasing the number of independent partners to the larger of the two finite numbers proves both assertions with one  $K_0$ . If no paired column is needed, one unused generic partner may be added so that  $K_0 > 0$ .  $\square$

## 8 The common space of spectral relations

The active space describes the possible responses at one block. We now identify the relations that remain when many independent blocks are used together. The essential point is that the derivatives at different triangular inputs belong to one common space. A bound on their ranks separately would not give the relations needed for the free-series construction.

We retain the preceding notation. The allowed sizes are  $\mathcal{S} = \{pN : N \geq 1\}$ ; at size  $n \in \mathcal{S}$ , there are  $e_n = dn/t$  distinct generic roots, each of multiplicity  $t$ . The active space has dimension  $a_n$ , and  $\rho_n$  is the generic rank of the eigenvalue differential, equivalently of the trace differential. In particular  $\rho_n \leq a_n$ . We use the replication identity  $a_{pN} = Na_p$  and the following consequence of the preceding attainability argument. There are a size  $r \in \mathcal{S}$  and an integer  $K_0 \geq 1$  such that the loop responses at a generic size- $r$  point, together with paired responses through  $K_0$  independent generic size- $r$  partners, span its active space. In each pairing, either arrow may be fixed generically and the opposite arrow varied. All spaces below are extended to a common splitting field when several independent points are involved.



**Lemma 8.1** (Derivatives at a star). *Let allowed-size blocks, evaluated by the preceding formal specialization conventions, have split semisimple outputs with disjoint spectra and the maximal total number of roots. Add input arrows only from designated hub blocks to designated leaf blocks, with no arrows between hubs or between leaves. The resulting input has the same roots and multiplicities as its diagonal part. A loop derivative at a block is its loop response  $D$ , with zero entries at all other blocks. If  $U$  is a forward arrow between a leaf  $P$  and a hub  $Q$ , a derivative in its reverse arrow  $V$  has entries*

$$(m_h(P, Q; U, V))_h, \quad (m_k(Q, P; V, U))_k$$

*at these two blocks and zero entries at the others.*

*Proof.* Order the leaves before the hubs. The input and its evaluated output are block upper triangular. The diagonal output blocks are the original ones; since their spectra are disjoint, the output is similar to their direct sum. In particular it is semisimple with the stated roots and multiplicities.

Differentiate  $\text{tr}(W^j)$ . A closed block path contributing to a reverse-arrow derivative must use that reverse arrow once and its matching forward arrow once. There is no other way to return to the starting block in this star. Its contribution is therefore exactly the mixed-arrow contribution at the diagonal input. Loop derivatives use no off-diagonal return. The previously proved mixed-response formula gives the displayed pairs after converting trace derivatives to root derivatives. This conversion is invertible on all distinct roots: its entries are  $jt\lambda_h^{j-1}$ , and the corresponding Vandermonde determinant is nonzero. The same reasoning applies to specialized blocks whenever the stated maximal-count and disjointness hypotheses hold.  $\square$

**Lemma 8.2** (A stable upper bound for the differential rank). *One can choose an arbitrarily large allowed size  $b$  and integers  $a = a_b$  and  $g \geq 0$ , such that*

$$\rho_b = a - g, \quad \rho_{bM} \leq Ma - g \quad (M \geq 1).$$

*At a generic size- $b$  point choose one root from each part of the active partition. Projection onto these  $a$  selected roots identifies the active space with the full coordinate space. The quotient of that coordinate space by the loop-response image has dimension  $g$ .*

*Proof.* For  $N \geq 1$ , put

$$g_N = Na_r - \rho_{rN}.$$

Replication and  $\rho_{rN} \leq a_{rN}$  show that  $g_N \geq 0$ . For  $N \geq K_0$ , take  $K_0$  of  $N$  independent size- $r$  blocks as hubs and the remainder as leaves. By Lemma 8.1 and the attainability property, the projection of the derivative image onto the leaf active spaces is surjective. Reverse-arrow variables belonging to different leaves are independent. Thus

$$\rho_{rN} \geq (N - K_0)a_r, \quad g_N \leq K_0a_r.$$

Here the specialized rank is bounded by the generic rank by the formal minor identities. The finitely many smaller values of  $N$  do not affect boundedness.

A bounded sequence of nonnegative integers has an eventual minimum that is attained arbitrarily late. Choose  $N_0 \geq K_0$  at such an occurrence, and write

$$g = g_{N_0}, \quad g_N \geq g \quad (N \geq N_0), \quad b = rN_0, \quad a = N_0a_r.$$

Then  $a = a_b$ ,  $\rho_b = a - g$ , and

$$\rho_{bM} = Ma - g_{N_0M} \leq Ma - g.$$

No eventual constancy of  $g_N$  is required. Finally, the active partition says that projection to one representative of each part is an isomorphism on the active space. The loop image has rank  $\rho_b$ , so its quotient has dimension  $g$ .  $\square$

Fix  $b, a, g$  as in Lemma 8.2. For each generic size- $b$  point  $P$ , let  $E_P$  be the  $a$ -dimensional space of selected-root coordinates, and let

$$L_P = \text{im } D_P \subseteq E_P, \quad H_P = E_P / L_P.$$

Thus  $\dim H_P = g$ . We regard  $H_P^*$  as the space of covectors on  $E_P$  annihilating all loop responses.

**Lemma 8.3** (Surjectivity at both ends of a pair). *Let  $P, Q$  be independent generic size- $b$  points. For a fixed generic arrow  $U$  between them, varying the reverse arrow  $V$  gives a joint paired-response map into  $H_P \oplus H_Q$  whose projection to either summand is surjective. This holds whichever orientation is fixed.*

*Proof.* Specialize  $P$  and  $Q$  to diagonal sums of  $N_0$  independent size- $r$  points. For each little block in  $P$ , the  $N_0 \geq K_0$  blocks in  $Q$ , together with their components of the fixed generic arrow, provide enough partners to span its active space after loop columns are included. The reverse-arrow entries touching different little blocks of  $P$  vary independently. Compression at a root of one little block sees only the reverse entries at that block. Consequently loops and paired columns span all  $N_0 a_r = a$  active coordinates at the  $P$ -end.

The resulting nonzero rank minor persists at generic  $P, Q, U$ . The labeled root data and the pair inverses are regular at the disjoint diagonal specialization, so this is precisely the determinant specialization justified earlier. Passing to the loop quotient gives surjectivity onto  $H_P$ . The same proof works at the  $Q$ -end and with the other orientation fixed. The finitely many nonzero minor conditions hold simultaneously on a nonempty generic set.  $\square$

The next lemma supplies the common subspace promised at the start of this section. Its formal-coordinate argument is the reason we obtain compatible relations between pairs of blocks.

**Lemma 8.4** (A common tangent space). *Fix  $M \geq 1$  independent generic size- $b$  points and their selected roots. At inputs with this diagonal and arbitrary strictly upper block arrows, the collective span of all selected-root derivative columns has dimension at most  $Ma - g$ . The span of all loop columns and all two-block paired columns has dimension exactly  $Ma - g$ .*

*Proof.* Put  $R = Ma - g$ . For  $M = 1$  the assertion is the definition of the loop rank, so suppose  $M \geq 2$ . Use one block as a hub, ordered last. The loops span  $\bigoplus_i L_{P_i}$ , of dimension  $M(a - g)$ . Modulo that space, reverse derivatives project surjectively onto each of the  $M - 1$  leaf quotients by Lemma 8.3. These projections vary independently. The derivative rank at this star is therefore at least

$$M(a - g) + (M - 1)g = R.$$

The generic upper bound in Lemma 8.2 is  $R$ . Thus at a generic strictly upper-arrow array  $\mathcal{U}$  the selected-root derivative rank equals  $R$ .

Work over the field containing  $\mathcal{U}$  and the diagonal splitting data. Let  $F(H)$  be the formal Taylor map of the selected root shifts under unrestricted input-entry perturbations  $H$ . The base

has the maximal root count, so the scalar branch construction applies. Every  $(R + 1)$ -minor of  $dF$  is zero, and some  $R$ -minor is a unit at  $H = 0$ . Choose the corresponding  $R$  output coordinates and use them, together with the remaining input coordinates, as new formal source coordinates  $(y, z)$ . The formal inverse theorem gives

$$F(y, z) = (y, G(y, z)).$$

The vanishing  $(R + 1)$ -minors now imply  $\partial G / \partial z = 0$ . In characteristic zero, coefficientwise differentiation gives  $G(y, z) = G(y)$ . At any formal source point where  $F = 0$ , all derivative columns therefore belong to the fixed graph

$$\{(v, dG(0)v) : v \in E^R\}.$$

Shifting only the strictly upper arrows leaves every root unchanged. Apply the preceding conclusion to these shifts: their Jacobian columns all belong to this one graph, not merely to spaces of the same dimension.

It remains to pass from formal shifts at  $\mathcal{U}$  to all upper-arrow arrays. Let  $J(U)$  be the selected Jacobian at such an array. It is polynomial in  $U$ . Indeed a trace derivative has just one differentiated input step, and every other off-diagonal step is strictly upper; a contributing block path therefore uses a bounded number of upper arrows. The remaining diagonal sums converge by smallness. The conversion from trace derivatives to root derivatives has coefficients depending only on the fixed diagonal roots, so it preserves polynomial dependence.

Every covector annihilating the preceding graph annihilates the polynomial matrix  $J(\mathcal{U} + S)$  as a formal identity in the upper-arrow shift  $S$ . Substitution  $S = V - \mathcal{U}$  is valid in this polynomial identity. Hence it annihilates every coefficient of  $J(V)$ . The span of those coefficients, formed over the original diagonal field, has dimension at most  $R$ , since scalar extension preserves rank. It contains every specialized column and proves the collective-span assertion.

Loop columns occur at the diagonal input. A paired column supported on two blocks occurs at the base with just the corresponding single upper arrow, by Lemma 8.1. Allowing both arrows to vary makes the collection independent of which orientation is called forward. All these columns therefore lie in the common space. The one-hub star already gives  $R$  independent columns among them, proving equality.  $\square$

**Proposition 8.5** (Compatible dual relations). *For any finite nonempty collection of independent generic size- $b$  points  $P_i$ , there is a  $g$ -dimensional space*

$$\mathcal{G}_{\{P_i\}} \subseteq \bigoplus_i H_{P_i}^*$$

*with the following properties.*

1. *Projection to each  $H_{P_i}^*$  is an isomorphism.*
2. *It annihilates all joint paired responses, with both arrows arbitrary.*
3. *For a fixed generic arrow between  $P_i, P_j$ , the annihilator in  $H_{P_i}^* \oplus H_{P_j}^*$  of the variable reverse-arrow image is exactly the restriction of  $\mathcal{G}_{\{P_i\}}$ .*
4. *On adjoining independent blocks, restriction of the larger space is the original space after scalar extension.*

*For one block this space is  $H_P^*$ . These statements include  $g = 0$ .*

*Proof.* Define  $\mathcal{G}$  as the annihilator of all loop and joint paired columns in the selected coordinate space. By Lemma 8.4, it has dimension  $g$ ; annihilation of loops identifies it with a subspace of

$\bigoplus_i H_{P_i}^*$ . If an element has zero component at one vertex  $i$ , its pairing with columns on  $(i, j)$  and the surjectivity onto  $H_{P_j}$  force its component at every other vertex  $j$  to vanish. Projection onto any one component is therefore injective, and hence an isomorphism by dimension.

For a fixed generic arrow on  $(i, j)$ , the reverse-image in  $H_{P_i} \oplus H_{P_j}$  has dimension at least  $g$ , since its projection onto either summand is surjective. It is killed by the  $g$ -dimensional restriction of  $\mathcal{G}$ . The latter's annihilator has dimension  $2g - g = g$ , so these two spaces coincide. This proves the exact pairwise assertion.

Finally, restriction from a larger collection annihilates every old loop and pair column. It is injective, because projection to any retained vertex is injective, and both spaces have dimension  $g$ . It is therefore onto. For a single block the same statement follows from its quotient dimension. All dimension arguments remain valid when  $g = 0$ .  $\square$

We henceforth write  $\mathcal{G}$  for these compatible spaces, specifying the blocks when needed. We have identified all linear relations shared by the loop and paired corrections. The next section proves that the leading free-word errors satisfy the corresponding cyclic relations.

## 9 Cyclic relations on the leading free-word errors

The space  $\mathcal{G}$  controls the linear corrections available at several blocks. We must show that the errors to be corrected satisfy its relations for individual cyclic orbits of free words. Commuting scalar substitutions would combine too many words. We instead substitute matrices, repeat each block, and attach hubs whose derivative columns identify the possible left nullspace.

Traces of words and their cyclic symmetry are classical in matrix invariant theory; see [16, §4]. Here the response relations must first be proved to annihilate every matrix trace test. An explicit directed-cycle calculation then isolates each cyclic orbit.

Fix  $m \geq 1$  independent generic size- $b$  points  $P_i$ . Write  $I_i$  for their selected root labels, so  $|I_i| = a$ , and let  $\mathcal{G} \subseteq \bigoplus_i H_{P_i}^*$  be the space of Proposition 8.5. As before, all scalars may be extended to a common field containing the required splitting data.

**Theorem 9.1** (Cyclic compatibility of the error). *Let  $Z(s_1, s_2)$  be a pair of formal series in two free letters, with constant term  $\text{diag}(P_1, \dots, P_m)$ . Suppose that the fixed-root Schur expressions  $S_{i,h}$ , for every block  $i$  and every root label  $h$ , vanish in every degree below  $\ell$ , where  $\ell \geq 1$ . By transfer, their degree- $\ell$  coefficients are scalar on their root spaces; write*

$$[s^\omega]S_{i,h} = e_{i,h,\omega}\pi_{i,h}, \quad |\omega| = \ell.$$

*For each word the scalar tuple at block  $i$  belongs to its active space. For every cyclic orbit  $\mathcal{O}$  of words of length  $\ell$  and every  $c \in \mathcal{G}$ ,*

$$\sum_{\omega \in \mathcal{O}} \sum_{i=1}^m \sum_{h \in I_i} c_{i,h} e_{i,h,\omega} = 0. \quad (9.1)$$

*The orbit sum runs over distinct words, including when a word has a nontrivial rotational stabilizer.*

We prove the theorem in three steps. First, arbitrary matrix substitutions turn the free-word errors into matrices on a repeat factor. Second, a derivative test shows that every relation in  $\mathcal{G}$  annihilates their traces. Finally, directed-cycle matrices isolate each cyclic orbit.

**Lemma 9.2** (Splitting repeated blocks). *Fix  $L \geq 1$  and arbitrary scalar matrices  $S_1, S_2 \in M_L(E)$ . Substitute  $s_c = \delta S_c$  into  $Z$ , with a commuting Taylor parameter  $\delta$ , and add an order- $\delta^\ell$  input loop  $Y_i$  in each old block  $i$ . Here  $Y_i$  is an arbitrary pair in  $M_b(E) \otimes M_L(E)$ . Put*

$$B_{i,h} = \sum_{|\omega|=\ell} e_{i,h,\omega} S^\omega + D_{i,h}(Y_i),$$

where  $D_{i,h}$  is applied entrywise to the repeat-factor coefficients. One can choose the  $Y_i$  so that every  $B_{i,h}$ , for all root labels  $h$ , has simple spectrum. For such a choice, after a finite splitting extension the leaf output has branches

$$\lambda_{i,h,j}(\delta) = \lambda_{i,h} + \delta^\ell \theta_{i,h,j} + O(\delta^{\ell+1}), \quad 1 \leq j \leq L,$$

each scalar of multiplicity  $t$ , where  $\theta_{i,h,j}$  are the eigenvalues of  $B_{i,h}$ . Their spectral projections are integral in  $\delta$  and reduce to  $\pi_{i,h} \otimes \Pi_{i,h,j}$ , where  $\Pi_{i,h,j}$  are the rank-one spectral projections of  $B_{i,h}$ .

*Proof.* Matrix substitution is legitimate coefficientwise in  $\delta$ . The leading Schur expression on the repeated  $(i, h)$ -space is  $I_t \otimes B_{i,h}$ : the original errors give the first sum, and an order- $\delta^\ell$  input loop contributes its first response. All lower orders are zero.

For each label  $h$ ,  $D_{i,h}$  is a nonzero linear functional on the input directions. Its matrix-valued extension is therefore onto  $M_L(E)$ . The discriminant of  $B_{i,h}$ , as a polynomial in the loop coordinates, is consequently not identically zero. There are finitely many labels, so the product of these discriminants is nonzero. Over the infinite field in use, or a scalar extension, choose loops at which their product is nonzero. This choice is made for all labels, not merely the selected ones. For  $L = 1$  the simple-spectrum condition is automatic.

Separate the old distinct roots by a formal conjugation equal to the identity at  $\delta = 0$ . The separated  $(i, h)$ -block is

$$\lambda_{i,h} I + \delta^\ell (I_t \otimes B_{i,h}) + O(\delta^{\ell+1}). \quad (9.2)$$

Here is an exact comparison with the fixed-root Schur expression. Write  $M = W - \lambda_{i,h} I$  in the fixed root/complement decomposition, with blocks  $M_{uv}$ . Its complementary block  $M_{22}$  is invertible. If  $F$  is the separated root block minus  $\lambda_{i,h} I$ , its invariant columns can be written  $\begin{pmatrix} Q \\ R \end{pmatrix}$ , with  $Q = I + O(\delta)$ ,  $R = O(\delta)$ , and  $M \begin{pmatrix} Q \\ R \end{pmatrix} = \begin{pmatrix} Q \\ R \end{pmatrix} F$ . Eliminating the lower row gives

$$(M_{11} - M_{12} M_{22}^{-1} M_{21}) Q = (Q - M_{12} M_{22}^{-1} R) F.$$

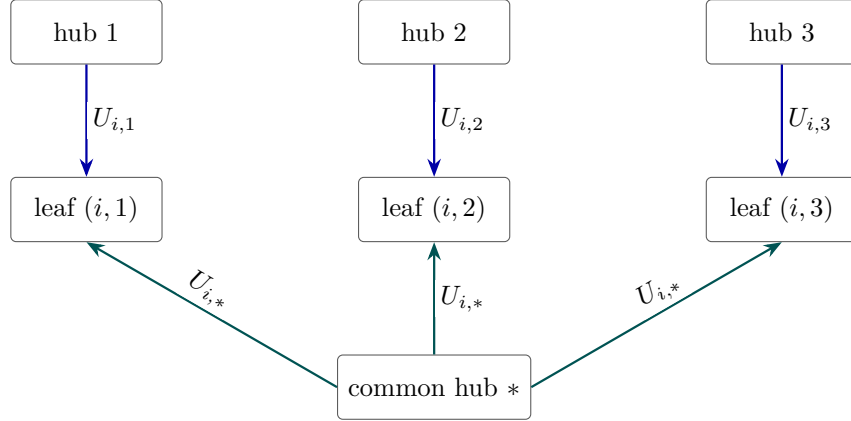
The left parenthesis is the fixed-root Schur expression. Both outer factors relating it to  $F$  reduce to identity at zero. Consequently it and  $F$  have the same first nonzero coefficient. Its vanishing below degree  $\ell$  forces the same vanishing for  $F$ , and its degree- $\ell$  coefficient is  $I_t \otimes B_{i,h}$ . This proves (9.2).

Within this block divide the change from  $\lambda_{i,h} I$  by  $\delta^\ell$ . Its constant term is  $I_t \otimes B_{i,h}$ , with  $L$  distinct eigenvalues. A second formal conjugation, integral in  $\delta$ , separates these slopes into blocks of size  $t$ . Across all old labels there are exactly

$$m L e_b = e_{bmL}$$

such blocks. Their characteristic polynomials are pairwise coprime over the Laurent fraction field: different old roots are already distinct at order zero, and distinct slopes separate the blocks inside one old root. The minimal-degree bound at size  $bmL$  is  $e_{bmL}$ . Each separated block contributes at least one to that degree, so each contributes exactly one and is scalar. The two integral conjugations also give integral spectral projections with the stated limits.  $\square$

One old block  $i$ , with three repeat positions at  $\delta = 0$



The common hub uses the same matrix pair at every repeat position.

Reverse arrows are zero at the base point and supply derivative directions.

Figure 2: The hub test at  $\delta = 0$ , for one old block  $i$  and  $L = 3$ . Hub  $p$  sends a generic matrix pair only to position  $p$ ; the common hub sends the same pair to all positions. Every old block uses these same  $L + 1$  hubs. At  $\delta \neq 0$  the leaf input may couple repeat positions. The individual hubs eliminate off-diagonal dual entries, and the common hub identifies the diagonal functionals.

Keep a choice as in Lemma 9.2. Adjoin  $L + 1$  independent generic size- $b$  hub blocks, indexed by  $p = 1, \dots, L$  and by  $*$ , with spectra disjoint from the old blocks. The individual hub  $p$  sends an input arrow to repeat position  $p$  of every old block  $i$ , with fixed generic value  $U_{i,p}$ . The common hub  $*$  sends the same generic value  $U_{i,*}$  to every repeat position in old block  $i$ . There are no reverse arrows at this test input. Arrow values for different pairs of old blocks and hubs can be chosen independently and to satisfy the surjectivity conditions of Lemma 8.3.

Figure 2 shows the distinction between the individual hubs and the common hub.

Treat the entire deformed leaf array as one block and order it before the hubs. Over the  $\delta$ -Laurent field this is a star with maximal root count. Put

$$N = mL + L + 1.$$

Select the  $L$  branches over each old selected label and the selected labels of all hubs, for a total of  $Na$  output rows. The rank of their derivative matrix is at most  $\rho_{bN} \leq Na - g$ . This bound follows from the original trace-minor identities, evaluated coefficientwise at the present Taylor input; it does not require the specialized input to be generic.

**Lemma 9.3** (The limiting derivative columns). *The following derivative columns at the preceding star have finite limits at  $\delta = 0$ .*

1. *A loop  $Y \otimes T$  inside old block  $i$  has limit*

$$D_{i,h}(Y) \operatorname{tr}(\Pi_{i,h,j} T)$$

*on row  $(i, h, j)$ , and zero on all other rows. Hub loops give their own loop responses.*

2. *For individual hub  $p$ , a reverse arrow of value  $V$  from repeat position  $r$  in old block  $i$  has limit*

$$\begin{aligned} m_{i,h}(P_i, Q_p; U_{i,p}, V) \operatorname{tr}(\Pi_{i,h,j} E_{pr}) & \quad \text{on row } (i, h, j), \\ \mathbf{1}_{p=r} m_{p,k}(Q_p, P_i; V, U_{i,p}) & \quad \text{on hub row } (p, k), \end{aligned}$$

*and zero elsewhere. Here  $E_{pr}$  is the repeat-factor matrix unit with row  $p$  and column  $r$ .*

3. *For the common hub, use arrow  $U_{i,*}$  and replace the repeat-factor matrix by*

$$\sum_{p=1}^L E_{pr}.$$

*The hub contribution is the single response  $m_{*,k}(Q_*, P_i; V, U_{i,*})$ .*

4. *The derivative along the whole input  $\delta$ -curve, multiplied by  $\delta^{1-\ell}$ , has limit  $\ell\theta_{i,h,j}$  on leaf rows and zero on hub rows.*

*The matrix consisting of a finite spanning collection of these limiting columns has rank at most  $Na - g$ .*

*Proof.* Over the Laurent field, apply Lemma 8.1 with the whole leaf array as one block. For loop derivatives, the bounded leaf projection tends to  $\pi_{i,h} \otimes \Pi_{i,h,j}$ . Compression of the ordinary first response gives  $D_{i,h}(Y)$  on the multiplicity space, and taking trace divided by  $t$  gives the first formula.

For paired derivatives on a leaf, the Schur inverse is on the hub at a root  $\lambda_{i,h,j}(\delta)$ ; that root stays away from the hub spectrum at  $\delta = 0$ . For paired derivatives on a hub, the inverse is on the leaf array at a hub root, which stays away from every old leaf root. These inverses are regular at  $\delta = 0$ . No inverse at the difference of two approaching leaf roots occurs in this calculation.

At the reduction, the leaf input is diagonal between old blocks and repeated inside each old block. A forward arrow at position  $p$  followed by a reverse arrow from position  $r$  inserts  $E_{pr}$  on the leaf repeat factor. At the hub end, the repeat indices must match, giving  $\mathbf{1}_{p=r}$ . The ordinary paired-response formula and trace divided by  $t$  now give the second formula. Summing forward positions gives the common-hub formula. The bounded projections and the distant-root inverses justify all these limits coefficientwise.

The curve column lies in the span of input derivative columns over the Laurent field by the chain rule. The expansion in Lemma 9.2 gives

$$\delta^{1-\ell} \frac{d}{d\delta} \lambda_{i,h,j}(\delta) = \ell\theta_{i,h,j} + O(\delta).$$

Hub roots are constant on the curve. Thus this column also has a finite limit. Choose bases for the finitely many loop and reverse direction spaces and include the curve column. Every  $(Na - g + 1)$ -minor of the resulting matrix vanishes over the Laurent field; since its entries have finite limits, these minors remain zero after specialization. This proves the last assertion.  $\square$



We now use the individual hubs to remove off-diagonal repeat-factor weights and the common hub to make all diagonal weights agree. The rank bound will then show that no element of  $\mathcal{G}$  has been lost.

**Lemma 9.4** (The left nullspace). *The left nullspace of the limiting matrix in Lemma 9.3, including its curve column, has dimension  $g$ . Its projection to the old blocks identifies it with  $\mathcal{G}$ : for each  $c \in \mathcal{G}$ , its leaf weights are  $c_{i,h}$  on every row  $(i, h, j)$ , independently of  $j$ , and its hub weights are the uniquely determined compatible functionals. Consequently*

$$\sum_i \sum_{h \in I_i} c_{i,h} \operatorname{tr}(B_{i,h}) = 0 \quad (c \in \mathcal{G}). \quad (9.3)$$

*Proof.* Let  $\mu_{i,h,j}$  be leaf row weights in the left nullspace and let  $\nu_p, \nu_*$  be the hub weights. Define

$$T_{i,h} = \sum_{j=1}^L \mu_{i,h,j} \Pi_{i,h,j} \in M_L(E).$$

The loop columns imply

$$\sum_{h \in I_i} D_{i,h}(Y) \operatorname{tr}(T_{i,h} T) = 0 \quad \text{for every } Y, T.$$

Thus, for every matrix entry  $(r, p)$ , the coordinate vector  $((T_{i,h})_{rp})_{h \in I_i}$  defines a functional in  $H_{P_i}^*$ . The hub weights similarly belong to their own loop annihilators.

Fix an individual hub  $p$  and a reverse position  $r \neq p$ . There is no hub response. Since

$$\operatorname{tr}(T_{i,h} E_{pr}) = (T_{i,h})_{rp},$$

the reverse columns say that the functional with these  $(r, p)$ -entries kills the leaf endpoint projection of the paired map. That projection is onto  $H_{P_i}$ , so all these entries vanish. As  $p, r$  vary, every  $T_{i,h}$  is diagonal. When  $r = p$ , the corresponding diagonal-entry functional and the hub functional  $\nu_p$  are exactly a pair in the relation of Proposition 8.5.

For the common hub, the leaf expression is

$$\operatorname{tr} \left( T_{i,h} \sum_p E_{pr} \right) = \sum_p (T_{i,h})_{rp} = (T_{i,h})_{rr}.$$

For every  $r$ , this diagonal-entry functional is therefore paired with the same  $\nu_*$ . Projection of the pair relation onto the hub dual space is an isomorphism, so the diagonal-entry functionals are independent of  $r$ . Compatibility for the collection of old blocks and hubs identifies them across all old blocks. Hence

$$T_{i,h} = c_{i,h} I_L, \quad c \in \mathcal{G},$$

and all hub weights are uniquely determined by  $c$ . Since the rank-one projections  $\Pi_{i,h,j}$  are mutually orthogonal and sum to identity, this also gives  $\mu_{i,h,j} = c_{i,h}$ . The map from the left nullspace to  $\mathcal{G}$  is consequently injective.

There are  $Na$  rows, and the limiting rank is at most  $Na - g$  even after the curve column is included. The nullspace therefore has dimension at least  $g$ . Its injection into the  $g$ -dimensional space  $\mathcal{G}$  is an isomorphism. In particular every  $c \in \mathcal{G}$ , not just a possible subspace of relations, occurs. Pairing its null vector with the curve column and cancelling the nonzero integer  $\ell$  yields

$$0 = \sum_i \sum_{h \in I_i} \sum_{j=1}^L c_{i,h} \theta_{i,h,j} = \sum_i \sum_{h \in I_i} c_{i,h} \operatorname{tr}(B_{i,h}).$$



If  $g = 0$ , the same proof gives zero nullspace and the assertion about  $c$  is vacuous. If  $L = 1$ , the off-diagonal step has no entries to remove; the common-hub step still gives precisely the stated conclusion.  $\square$

*Proof of Theorem 9.1.* Apply Lemma 9.4 to arbitrary  $S_1, S_2$  and to the auxiliary loops chosen in Lemma 9.2. The added-loop contribution to (9.3) is zero. Indeed, write  $Y_i = \sum_{\alpha} Y_{i,\alpha} \otimes T_{i,\alpha}$ . Then

$$\sum_{h \in I_i} c_{i,h} \operatorname{tr}(D_{i,h}(Y_i)) = \sum_{\alpha} \operatorname{tr}(T_{i,\alpha}) \sum_{h \in I_i} c_{i,h} D_{i,h}(Y_{i,\alpha}) = 0,$$

because  $c_i$  annihilates the loop image. We conclude that, for arbitrary matrix substitutions,

$$\sum_{|\omega|=\ell} \left( \sum_i \sum_{h \in I_i} c_{i,h} e_{i,h,\omega} \right) \operatorname{tr}(S^{\omega}) = 0. \quad (9.4)$$

Auxiliary field extensions impose no restriction: this equality has its coefficients in the original field, and extension is injective.

We use a matrix-cycle test of the kind appearing in the constructive trace argument of [11, Proposition 4.1], recording the rotational multiplicity explicitly. Fix a word  $\omega_0 = c_0 c_1 \cdots c_{\ell-1}$ , with letters  $c_j \in \{1, 2\}$ , and take  $L = \ell$ . On vertices indexed modulo  $\ell$ , set

$$S_c = \sum_{j: c_j=c} E_{j,j+1} \quad (c = 1, 2).$$

For a word  $\omega$  of length  $\ell$ , its matrix trace counts the starting vertices at which one reads  $\omega$  around this directed cycle. It is zero unless  $\omega$  lies in the cyclic orbit  $\mathcal{O}$  of  $\omega_0$ ; for every distinct word in that orbit its value is the same positive integer

$$\frac{\ell}{|\mathcal{O}|}.$$

This is the rotational stabilizer size and includes periodic and constant-letter words. Substitution into (9.4), followed by cancellation of that integer in characteristic zero, gives (9.1).  $\square$

The leading errors now satisfy every cyclic relation imposed by  $\mathcal{G}$ . The colored correction graph will make these exactly the annihilators of its order-by-order correction map.

## 10 Solving the free-series equations

We now use the response and cyclic-sum results to keep the output of a substitution conjugate to its constant term. The input retains two prescribed linear coefficients. This is the point at which free words, rather than commuting parameters, enter the construction essentially.

Work over a common split scalar field  $E$  in which the preceding evaluations and spectral projections are defined. All series substitutions below have small constant terms and are interpreted by the coefficientwise valuation convergence established earlier. In particular, the letters of a free series commute with its matrix coefficients; the letters do not commute with each other.

Put  $m = 11$  and index the blocks by  $i \in \mathbb{Z}/m\mathbb{Z}$ . Let  $P_i$  be jointly independent small generic pairs of  $b \times b$  matrices. Set  $R = \operatorname{diag}(P_i)$ , a pair of  $mb \times mb$  matrices. The old coefficient algebra acts on a separate factor of dimension  $d$ . Write  $\lambda_{i,h}$  and  $\pi_{i,h}$  for the distinct eigenvalues and spectral projections of  $w(P_i)$ . Eigenvalues belonging to different blocks are distinct. For a substitution  $Z$

with constant term  $R$ , let  $S_{i,h}(Z)$  denote the fixed-root Schur expression at  $\lambda_{i,h}$ , using the constant projection  $\pi_{i,h}$  in the  $i$ -th block.

We recall precisely the response information needed here. In each block choose one eigenvalue label from every part of the active partition, giving  $a$  selected labels. The selected loop-response map is

$$D_i : M_b(E)^2 \longrightarrow E^a.$$

Put  $L_i = \text{im } D_i$  and  $H_i = E^a/L_i$ . Proposition 8.5 supplies a vector space  $G = \mathcal{G}$  of dimension  $g$  and coordinate isomorphisms

$$p_i : G \longrightarrow H_i^*.$$

For a generic fixed input arrow  $U$  on  $i \leftarrow j$ , varying its opposite arrow  $V$  gives two selected responses

$$m_i(U, V) \in E^a, \quad m_j(V, U) \in E^a.$$

The annihilator of these paired responses in  $H_i^* \oplus H_j^*$  is exactly

$$\{(p_i(c), p_j(c)) : c \in G\}. \quad (10.1)$$

The identifications  $p_i$  are compatible for the whole collection of blocks and for all the generic arrows used below.

The transfer result supplies the following additional facts. If all Schur expressions vanish in degrees less than  $\ell$ , their degree- $\ell$  coefficients are scalar on the projected eigenspaces. The resulting tuples belong to their active spaces and are determined by their selected entries. Denote the selected error for a word  $\omega$  of length  $\ell$  in block  $i$  by  $e_{i,\omega} \in E^a$ . The cyclic-sum result says that, for every orbit  $\mathcal{O}$  of words under cyclic rotation,

$$\sum_{\omega \in \mathcal{O}} \sum_i p_i(c)(e_{i,\omega} + L_i) = 0 \quad (c \in G). \quad (10.2)$$

These statements hold for every partial substitution under consideration, including after an extension of its scalar field. The preceding results therefore supply all the hypotheses used in the next construction.

**Proposition 10.1.** *Prescribe generic input-arrow pairs on the edges*

| letter | offsets $i - j$ of edges $i \leftarrow j$ |
|--------|-------------------------------------------|
| $s_1$  | 1, 2                                      |
| $s_2$  | 3, 4.                                     |

*Under the response and cyclic-sum hypotheses above, there is a pair  $Z(s_1, s_2) \in M_{mb}(E) \langle\langle s_1, s_2 \rangle\rangle^2$  with constant term  $R$  and exactly these prescribed linear coefficients, for which*

$$S_{i,h}(Z) = 0 \quad \text{for every } i, h.$$

*It can be constructed by finite linear systems. At step  $\ell \geq 3$  the new unknowns are coefficients of degree  $\ell - 1$  on the reverse edges and coefficients of degree  $\ell$  on the diagonal blocks.*

*Proof.* None of the four forward offsets is zero. The sum of two forward offsets lies between 2 and 8, and is therefore nonzero modulo 11. Thus a path of one or two forward steps cannot return to its initial block. Constant input steps preserve each block. The path expansions for the output

and for its Schur expressions consequently show that the prescribed constant and linear terms, with zero higher coefficients, solve all the Schur equations through degree two.

Suppose a partial substitution solves them in degrees less than  $\ell \geq 3$ . Add a correction  $VB$  on a reverse edge  $j \leftarrow i$ , where  $B$  is a word of length  $\ell - 1$ , and add arbitrary degree- $\ell$  loop corrections. At the block-diagonal constant term, an off-diagonal input variation gives an output variation in the same off-diagonal block. Its compression on any fixed root block is zero. Thus the reverse correction changes no Schur equation through degree  $\ell - 1$ .

Its contribution in degree  $\ell$  must pair it with a single prescribed linear arrow. There is a unique forward edge that returns along this reverse edge: the reverse offsets 10, 9, 8, 7 correspond respectively to 1, 2, 3, 4. If that edge has coefficient  $U$  and color  $s_c$ , the contributions are

| block | word    | selected response |
|-------|---------|-------------------|
| $i$   | $s_c B$ | $m_i(U, V)$       |
| $j$   | $B s_c$ | $m_j(V, U)$ .     |

(10.3)

This is exactly the two-arrow Schur calculation defining the paired responses. A previously present coefficient of degree at least two cannot interact with the new degree- $\ell - 1$  coefficient in degree  $\ell$ . Two new corrections have total degree at least  $2(\ell - 1) > \ell$ . The degree- $\ell$  variation is therefore linear. Loop corrections give  $D_i$  independently at every vertex and every word of length  $\ell$ .

We compute the annihilator of this finite correction map. Let  $f_{i,\omega} \in (E^a)^*$  be a collection of functionals annihilating its image. The loop columns first imply that  $f_{i,\omega}$  annihilates  $L_i$ . Write its induced functional on  $H_i$  uniquely as

$$f_{i,\omega} = p_i(c_{i,\omega}), \quad c_{i,\omega} \in G.$$

Here and below we use the same notation for the induced functional and its pullback to  $E^a$ . By (10.1) and (10.3), annihilation of the reverse columns is equivalent to

$$c_{i,s_c B} = c_{j,B s_c} \quad \text{on every edge } i \leftarrow j \text{ of color } c. \quad (10.4)$$

The two offsets of either color are consecutive integers, say  $a, a + 1$ . Applying (10.4) to both edges out of  $j$  gives

$$c_{j+a,s_c B} = c_{j+a+1,s_c B}.$$

As  $j$  ranges over  $\mathbb{Z}/11\mathbb{Z}$ , this makes the value independent of the vertex. Equation (10.4) then identifies its value with that on the rotated word  $B s_c$ . Iterating rotations shows that the annihilator consists exactly of one element of  $G$  per cyclic orbit of words. The converse follows from the same paired-annihilator identity.

Equation (10.2) says that every such annihilator kills the degree- $\ell$  error. Finite-dimensional linear duality therefore puts the negative error in the image of the correction map. Choose a correction solving all selected equations. Lower degrees remain solved, so transfer and the active partition imply that every nonselected equation vanishes in degree  $\ell$  as well.

Continue inductively. A coefficient of degree  $n$  can change only at steps  $n$  and  $n + 1$ , so the resulting free series is well-defined. Every fixed Schur coefficient is eventually zero and remains zero. The construction also covers  $g = 0$ , when the annihilator is zero. For a periodic word the orbit consists of its distinct rotations; no multiplicity is inserted into the annihilator calculation or into (10.2).  $\square$

Figure 3 separates the generating cycle from the full local arrow pattern. The role of the four offsets is now explicit. Avoiding returns of length one or two starts the induction without changing the linear data. Consecutive offsets within each color make its only obstructions independent of the vertex. The remaining obstructions are exactly the cyclic sums already proved to vanish.

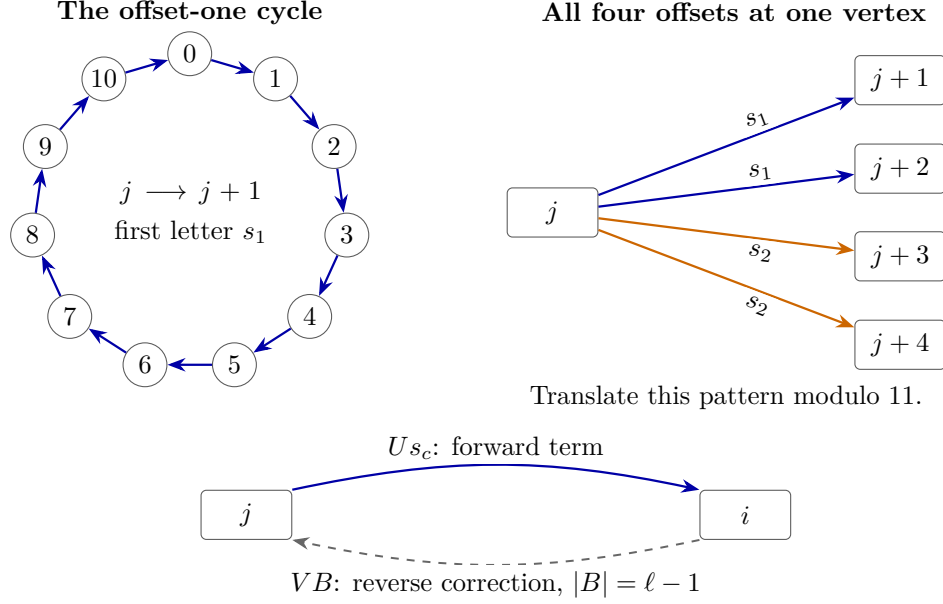


Figure 3: The eleven-block input pattern. Indices lie in  $\mathbb{Z}/11\mathbb{Z}$ . The left panel shows the offset-one cycle; the right panel specifies all four forward offsets, translated at every vertex. Each forward arrow carries a generic pair of input matrices. At order  $\ell \geq 3$ , a reverse correction has degree  $\ell - 1$ ; degree- $\ell$  loop corrections are not drawn.

## 10.1 Fixed roots and Schur complements

The following elementary fact identifies the spectral equations with a single polynomial identity. Its coefficient ring need not be commutative. Let  $E$  be a field, let  $\mathfrak{a} = (s_1, s_2)$  be the augmentation ideal of the free algebra  $E\langle s_1, s_2 \rangle$ , and put

$$T_N = E\langle s_1, s_2 \rangle / \mathfrak{a}^{N+1} \quad (N \geq 0).$$

Elements of  $E$  are central in  $T_N$ . Reduction modulo  $\mathfrak{a}$  is called the constant term.

**Lemma 10.2.** *Let  $W \in M_d(T_N)$  have constant term*

$$W_0 = \sum_{h=1}^e \lambda_h \pi_h, \quad \pi_h \pi_j = \delta_{hj} \pi_h, \quad \sum_{h=1}^e \pi_h = I_d,$$

where the  $\lambda_h \in E$  are distinct and the  $\pi_h \in M_d(E)$  are nonzero projections. Set

$$q(t) = \prod_{h=1}^e (t - \lambda_h), \quad \bar{\pi}_h = I_d - \pi_h.$$

On the complementary summand  $\bar{\pi}_h T_N^d$ , the operator

$$C_h = \bar{\pi}_h (W - \lambda_h I_d) \bar{\pi}_h$$

is invertible. Define the Schur expression on  $\pi_h T_N^d$  by

$$S_h = \pi_h (W - \lambda_h I_d) \pi_h - \pi_h W \bar{\pi}_h C_h^{-1} \bar{\pi}_h W \pi_h.$$

Then  $q(W) = 0$  if and only if  $S_h = 0$  for every  $h$ .

*Proof.* The constant term of  $C_h$  is  $\sum_{j \neq h} (\lambda_j - \lambda_h) \pi_j$ , which is invertible on  $\bar{\pi}_h E^d$ . A matrix over  $T_N$  with invertible constant term is invertible: after factoring out its constant term, the inverse is a finite geometric series in a matrix with entries in  $\mathfrak{a}$ . This proves the assertion about  $C_h$ . All modules below are right  $T_N$ -modules, with matrices acting on column vectors from the left.

Suppose first that every  $S_h$  vanishes. Define  $K_h : \pi_h T_N^d \rightarrow T_N^d$  by

$$K_h u = u - C_h^{-1} \bar{\pi}_h W \pi_h u,$$

where the second term lies in the complementary summand. Then  $\pi_h K_h u = u$ , and direct block multiplication gives

$$(W - \lambda_h I_d) K_h u = S_h u = 0.$$

The map  $K = \sum_h K_h \pi_h$  has constant term  $I_d$ , so it is invertible. The displayed kernel equations imply  $WK = KW_0$ . Consequently

$$q(W) = Kq(W_0)K^{-1} = 0.$$

Conversely, suppose  $q(W) = 0$ . Define

$$P_h = \prod_{j \neq h} \frac{W - \lambda_j I_d}{\lambda_h - \lambda_j}.$$

These products are unambiguous because their factors are polynomials in the same matrix with central scalar coefficients. Polynomial interpolation modulo  $q$  shows that the  $P_h$  are spectral idempotents; in particular,

$$(W - \lambda_h I_d) P_h = 0, \quad P_h \equiv \pi_h \pmod{\mathfrak{a}}.$$

Let  $\iota_h : \pi_h T_N^d \rightarrow T_N^d$  be inclusion. The endomorphism  $A_h = \pi_h P_h \iota_h$  of  $\pi_h T_N^d$  has constant term the identity, and is therefore invertible. Thus

$$\widetilde{K}_h = P_h \iota_h A_h^{-1}$$

satisfies  $\pi_h \widetilde{K}_h = \text{id}_{\pi_h T_N^d}$  and  $(W - \lambda_h I_d) \widetilde{K}_h = 0$ . The complementary component of the latter equation determines its remaining component uniquely:

$$\bar{\pi}_h \widetilde{K}_h = -C_h^{-1} \bar{\pi}_h W \pi_h.$$

Its  $\pi_h$  component now gives  $S_h = 0$ . □

For a matrix over  $E\langle\langle s_1, s_2 \rangle\rangle$ , apply Lemma 10.2 after reduction to  $T_N$ . It follows that the fixed-root Schur equations hold through degree  $N$  exactly when the identity  $q(W) = 0$  holds through degree  $N$ . In particular, this equivalence applies at every finite stage of a formal correction, with the same polynomial  $q$  throughout.

## 11 Realization in a central division algebra

The preceding construction solved equations in split matrices. We now place its constant blocks and prescribed arrows in one central division algebra. Each subsequent correction is the solution of a finite linear system over its center. This permits descent without any assertion that solutions of nonlinear equations descend.

Throughout this section  $k$  has characteristic zero and contains all roots of unity, and  $A$  is a central division algebra of degree  $d$  over  $k$ . The integer  $b$  is the block size chosen earlier in the spectral case. In the case where  $q_0(w) = 0$  already holds as a free-series identity, any positive integer  $b$  may be used. Put  $m = 11$ .

### 11.1 The two division-algebra factors

Choose independent central indeterminates  $\alpha, \beta$  and put  $k_0 = k(\alpha, \beta)$ . Let  $B_b$  be the symbol algebra over  $k_0$  generated by  $u, v$  with

$$u^b = \alpha, \quad v^b = \beta, \quad vu = \zeta_b uv,$$

where  $\zeta_b$  is a primitive  $b$ -th root of unity. For  $b = 1$  this means  $B_1 = k_0$ . By the construction in Section 4.2,  $B_b$  is central simple of degree  $b$ . Lemma 4.3 shows that  $A \otimes_k B_b$  is division and remains division under pure rational and central Laurent scalar extensions. We will use these facts twice in adjoining the cyclic block factor.

Fix a  $k_0$ -basis  $(e_a)_{a=1}^{b^2}$  of  $B_b$ . Over  $k_0$  introduce the following jointly independent variables:

$$x_{c,a,j}, \quad y_{r,c,a,j}, \quad \begin{array}{l} c \in \{1, 2\}, \quad 1 \leq a \leq b^2, \\ j \in \mathbb{Z}/m\mathbb{Z}, \quad r \in \{1, 2, 3, 4\}. \end{array}$$

Let  $L_0$  be their rational function field and let  $\sigma$  increase the index  $j$  by one, fixing  $k_0$ . This automorphism has order exactly  $m$ . Put  $H_0 = L_0^{\langle \sigma \rangle}$ . Artin's fixed-field theorem for a finite group of automorphisms gives a cyclic Galois extension  $L_0/H_0$  of degree  $m$ ; see [15, Corollary 3.11, p. 38].

Choose a further independent central indeterminate  $a_*$  and form the cyclic algebra  $C$  over  $F_0 = H_0(a_*)$  with coefficient field  $L_0(a_*)$  and generator  $\tau$ , subject to

$$\tau^m = a_*, \quad \tau c \tau^{-1} = \sigma(c) \quad (c \in L_0). \quad (11.1)$$

It has vector-space decomposition  $C = \bigoplus_{r=0}^{m-1} L_0(a_*) \tau^r$ . After extending scalars to a field containing  $L_0(a_*)$ , represent  $c \in L_0$  by the diagonal with entries  $\sigma^{-i}(c)$  and  $\tau$  by the cyclic forward shift, whose wrap entry is  $a_*$ . These matrices satisfy (11.1) and span all matrix units. This gives the usual splitting of  $C$  and shows that it is central simple of degree  $m$ ; see [4, Construction 2.5.1 and Proposition 2.5.2]. Our conjugation convention uses the inverse of their Galois generator.

Define

$$\tilde{B}_0 = (B_b \otimes_{k_0} F_0) \otimes_{F_0} C, \quad D_0 = A \otimes_k \tilde{B}_0.$$

Then  $D_0$  is central simple of degree  $dbm$  over  $F_0$  by the tensor-product and scalar-extension properties of central simple algebras [14, Chapter IV, Corollary 2.8 and Proposition 2.15]. It is also division. Indeed

$$D_L = (A \otimes_k B_b) \otimes_{k_0} L_0$$

is division by the rational-extension argument. Extend  $\sigma$  to  $D_L$  by fixing the old algebra factors. The skew Laurent division ring  $D_L((t; \sigma))$  receives a unital homomorphism from  $D_0$  by  $\tau \mapsto t$  and  $a_* \mapsto t^m$ . The central element  $t^m$  is transcendental over  $H_0$ . Simplicity again makes this map injective, and finite dimension makes  $D_0$  division.

Finally put

$$F = F_0((\varepsilon)), \quad \tilde{B} = \tilde{B}_0 \otimes_{F_0} F, \quad D = D_0 \otimes_{F_0} F.$$

The central Laurent argument shows that  $D$  is a central division algebra over  $F$  of degree  $dbm$ . Input coefficients will lie in  $\tilde{B}$ , so that they commute with the old factor  $A$ .

### 11.2 One realization of all generic data

For  $r \in \mathbb{Z}/m\mathbb{Z}$ , let

$$\mathcal{V}_r = (B_b \otimes_{k_0} L_0) \tau^r \otimes_{H_0} F \subseteq \tilde{B}.$$

This notation uses the evident coefficient embedding and scalar extension; equivalently it is the  $F$ -span of that graded summand in  $\tilde{B}_0$ . Upon splitting,  $\mathcal{V}_r$  consists exactly of all  $b \times b$  entries at offset  $r$  in the  $m$ -block matrix. In fact  $L_0 \otimes_{H_0} E \simeq E^m$  over a splitting field  $E$ , and  $B_b$  becomes  $M_b(E)$ . Multiplication by  $\tau^r$  moves these independent diagonal entries to offset  $r$ , with nonzero wrap factors. In particular  $\mathcal{V}_0$  gives all diagonal blocks, and  $\mathcal{V}_{-r}$  gives all reverse entries for offset  $r$ .

Define the two constant inputs and four arrow pairs by

$$P_c^0 = \varepsilon \sum_a x_{c,a,0} e_a, \quad U_{r,c} = \left( \sum_a y_{r,c,a,0} e_a \right) \tau^r \quad (c = 1, 2).$$

The prescribed initial substitution is the pair

$$Z_c^{\text{init}}(s_1, s_2) = P_c^0 + (U_{1,c} + U_{2,c})s_1 + (U_{3,c} + U_{4,c})s_2. \quad (11.2)$$

After splitting, the diagonal blocks of  $(P_1^0, P_2^0)$  are exactly the required independent small generic pairs. The two input-component entries at every forward edge are independent generic entries, jointly with all the diagonal data.

Here independence is over the old split coefficient field, which is the field used in the generic rank tests. To check it, first split  $A$  and  $B_b$  using constants independent of the variables  $x, y$ . The  $m$  conjugates of each coefficient use the disjoint sets of variables with different index  $j$ . The split symbol basis is a matrix basis. Permuting those variables, changing that basis, and multiplying entries by powers of  $a_*$  preserve their algebraic independence. This verifies all generic determinant conditions simultaneously. No independence over the larger invariant field  $H_0$  is asserted or needed.

We may perform the spectral calculations after extending  $F$  to a split Laurent field and then to the finite extensions needed for eigenvalues. Its constant coefficient field contains the old algebraic closure and the rational variables just chosen. The matrix constant terms remain  $\varepsilon$ -small. Arrow coefficients need not be small, since they occur only with positive free-word degree.

### 11.3 Linear descent of the correction

Assume first that  $q_0(w) \neq 0$ . Let  $W_0 = w(P_1^0, P_2^0) \in D$  and choose its monic minimal polynomial

$$q(t) \in F[t]. \quad (11.3)$$

We keep this polynomial fixed throughout the correction. It is irreducible, because  $F[W_0]$  is a finite-dimensional commutative domain, and separable because the characteristic is zero. In a splitting field its roots are exactly the distinct eigenvalues of the constant output. For completeness, left multiplication by  $W_0$  on  $D$  has characteristic polynomial a power of  $q$ , since  $D$  is a vector space over  $F[W_0]$ . After splitting, that left multiplication is a direct sum of copies of the action of the split matrix  $W_0$ . Every root of  $q$  therefore occurs in that matrix, and no other eigenvalue occurs. Separability makes its action semisimple.

We solve  $q(w(Z)) = 0$  in free series over  $D$ . Initially take (11.2). Its split Schur equations vanish through degree two, so Lemma 10.2 gives the same assertion for the polynomial relation. Suppose inductively that a partial substitution solves this relation through degree  $\ell - 1$ , where  $\ell \geq 3$ . Allow a new coefficient in  $\mathcal{V}_{-r}^2$  at every word of length  $\ell - 1$ , for  $r = 1, 2, 3, 4$ , and a new coefficient in  $\mathcal{V}_0^2$  at every word of length  $\ell$ . These are finite-dimensional  $F$ -spaces.

Impose every coefficient equation of

$$q(w(Z)) = 0 \pmod{(s_1, s_2)^{\ell+1}}. \quad (11.4)$$



This is a finite affine linear system over  $F$  in the allowed new coordinates. Indeed two new insertions have degree at least  $2(\ell - 1) > \ell$ . At any fixed free-word degree, only boundedly many positive-degree insertions occur in the evaluation of the old series; the remaining insertions are from the  $\varepsilon$ -small constant input. The finitely many previously fixed coefficients have a finite lower valuation bound. Thus the sums defining each coefficient of (11.4), and its linear part in the new coordinates, converge in  $D$  and are defined over  $F$ . Choosing an  $F$ -basis of  $D$  turns them into finitely many ordinary scalar linear equations.

After scalar extension, each permitted input space becomes exactly the collection of reverse edges or loops used in Proposition 10.1. The current partial substitution is a valid partial split solution by Lemma 10.2. The proposition supplies a split correction, and that lemma converts its Schur vanishings back to all the equations in (11.4). In particular the new degree- $\ell - 1$  reverse coefficients preserve the already solved equations: their diagonal Schur variation at that degree is zero. We have therefore solved the whole finite linear system after a field extension. A matrix and its augmented matrix have the same ranks before and after extension, so the system already has a solution over  $F$ .

Choose such a solution at every step. The coefficient stabilization in Proposition 10.1 produces

$$Z \in \tilde{B}\langle\langle s_1, s_2 \rangle\rangle^2, \quad q(w(Z)) = 0,$$

with the prescribed constant and linear terms. Only finite linear systems have been descended. Neither the roots of  $q$  nor the split correction coefficients need belong to  $F$ .

Since the input coefficients commute with  $A$ , evaluation at  $Z$  defines a substitution homomorphism

$$A\langle\langle z_1, z_2 \rangle\rangle \longrightarrow D\langle\langle s_1, s_2 \rangle\rangle.$$

It is well-defined by the same coefficientwise valuation bound: at fixed output degree, only boundedly many nonconstant input factors occur. The bound also justifies multiplication and regrouping of the convergent coefficient sums. Hence every old polynomial identity on the tuple and the chosen series is preserved by this substitution.

If  $q_0(w) = 0$  already holds, use the very same division algebra and the initial substitution (11.2), with any  $b \geq 1$ . No spectral correction is required; the old relation  $q_0(w)$  remains zero under the substitution homomorphism. Both cases produce a central division algebra of degree  $11bd > d$ , with identical generic forms for the constant and the two linear input coefficients. The next step restores generation and two transverse directions using these coefficients. That step is needed in the unconstrained case as well as in the spectral case.

## 12 Restoring generation and two transverse directions

The preceding construction gives a substitution  $Z(s_1, s_2)$  inside the centralizer of  $A$  in a larger central division algebra. It preserves the old polynomial identities and algebraizes the chosen series. To complete Theorem 2.1, we must make the new constant tuple generate the larger algebra and retain two independent directions modulo inner derivations. We first use the original transverse directions to prove generation of the generic diagonal product and to control the arrows between its blocks. Translating the first free letter by a new central parameter then joins the blocks into a full matrix algebra. A single rank calculation shows that both directions remain independent modulo inner derivations.

Throughout this section, brackets with a tuple are taken componentwise. Thus, for  $C = (C_1, \dots, C_u)$  in an algebra  $B$ , write

$$J(C) = \{([T, C_1], \dots, [T, C_u]) : T \in B\}.$$

Polynomial relations on  $X_0$  detect the two input directions to first order.

**Lemma 12.1** (Derivatives of relations). *Let  $A$  be a finite-dimensional central simple algebra over a field  $k$  of characteristic zero. Let  $X \in A\langle\langle z_1, z_2 \rangle\rangle^u$  have constant tuple  $X_0$  generating  $A$ , and suppose its two linear coefficients  $X_1, X_2$  are independent modulo  $J(X_0)$ . Define*

$$\theta : k\langle x_1, \dots, x_u \rangle \longrightarrow A, \quad \theta(f) = f(X_0), \quad I = \ker \theta.$$

*Then the map taking  $f \in I$  to the coefficients of  $z_1, z_2$  in  $f(X)$  is onto  $A^2$ .*

*Proof.* Denote these two coefficient maps by  $\delta_1, \delta_2$ . Each satisfies

$$\delta_j(fg) = \theta(f)\delta_j(g) + \delta_j(f)\theta(g).$$

Since  $\theta$  is onto and  $I$  is a two-sided ideal, the image  $M = (\delta_1, \delta_2)(I)$  is an  $A$ -bimodule in  $A^2$ .

Left and right multiplication generate  $\text{End}_k(A)$ ; see [14, Chapter IV, Corollary 2.9]. Indeed, after splitting  $A$ , multiplication by matrix units on the left and right gives every matrix unit on the vector space  $A$ ; the resulting isomorphism  $A \otimes_k A^{\text{op}} \simeq \text{End}_k(A)$  descends by finite-dimensional linear algebra. Applying these endomorphism matrix units to  $M \subseteq A \otimes_k k^2$  shows that  $M = A \otimes_k V$  for a subspace  $V \subseteq k^2$ . If  $V \neq k^2$ , some nonzero  $(a, b) \in k^2$  annihilates  $V$ . The map  $a\delta_1 + b\delta_2$  then vanishes on  $I$  and descends to a derivation of  $A$ .

Every  $k$ -linear derivation of  $A$  is inner [8, Theorem 8]. For completeness, after splitting  $A = M_d(E)$ , a derivation  $\partial$  is implemented by

$$T = \sum_i \partial(E_{i1})E_{1i}.$$

Differentiating  $E_{kl}E_{i1} = \delta_{li}E_{k1}$ , multiplying by  $E_{1i}$ , and summing gives  $\partial(E_{kl}) = [T, E_{kl}]$ . The equations for an implementing element on a fixed  $k$ -basis of  $A$  form a linear system over  $k$ . Solvability after scalar extension implies solvability over  $k$ . It follows that  $aX_1 + bX_2 \in J(X_0)$ , a contradiction.  $\square$

In particular, there are  $f_1, f_2 \in I$  whose derivative pairs are  $(1, 0)$  and  $(0, 1)$ . These relations recover the two generic input matrices from small evaluations of the tuple.

**Lemma 12.2** (Generic blocks). *Under the hypotheses of Lemma 12.1, write  $d = \deg A$  and split  $A$  into  $d \times d$  matrices. Fix a positive integer  $b$  and finitely many independent small generic input pairs*

$$P_i = \varepsilon P_i^\circ, \quad 1 \leq i \leq r,$$

*of  $b \times b$  matrices, with jointly independent entries in the  $P_i^\circ$ . Work over a common splitting Laurent field. Then each tuple  $C_i = X(P_i)$  generates  $M_{db}$ , and the diagonal tuple  $C = \text{diag}(C_i)$  generates the full product of these matrix algebras.*

*For distinct  $i, j$ , the output derivative of a generic input arrow  $U = (U_1, U_2)$  from block  $j$  to block  $i$  is nonzero modulo*

$$\{C_i T - T C_j : T \in M_{db}\}.$$

*More precisely, the image of the arrow derivatives in this quotient has dimension at least  $b^2$ .*

*Proof.* Let  $E$  contain the splitting data and independent input coordinates. Consider integral elements of the  $E((\varepsilon))$ -algebra generated by  $X(P_i)$ . Their reductions modulo  $\varepsilon$  contain  $M_d(E) \otimes 1$ , because  $X_0$  generates the old algebra. Lemma 12.1 also gives

$$\varepsilon^{-1} f_j(X(P_i)) \equiv I_d \otimes P_{i,j}^\circ \pmod{\varepsilon}.$$

A generic pair generates  $M_b(E)$ . To see the relevant determinant is nonzero, specialize one matrix to a diagonal matrix with distinct entries and the other to a matrix with every entry nonzero. Interpolation and sandwiching give all matrix units. Thus the reductions generate  $M_d(E) \otimes M_b(E) = M_{db}(E)$ . Choose finitely many integral algebra elements whose reductions form a matrix basis. Their determinant is a unit, so they remain independent over  $E((\varepsilon))$ , proving full generation.

The representations defined by different  $C_i$  have distinct kernels. Indeed,

$$\mathrm{tr} f_1(C_i) = \varepsilon d \mathrm{tr} P_{i,1}^\circ + O(\varepsilon^2),$$

and the leading coefficients differ. Equal kernels would induce an isomorphism of the full matrix images carrying every polynomial evaluation in one to the other. Such an isomorphism preserves trace: the trace of left multiplication is matrix size times matrix trace. This contradicts the display. The kernels are therefore distinct maximal two-sided ideals. Evaluation is onto their finite product. Explicitly, for each pair  $i \neq j$  choose a polynomial vanishing in block  $j$  and equal to the identity in block  $i$ . The product over  $j \neq i$  vanishes in all other blocks and is the identity in block  $i$ ; multiplying by a lift of any matrix gives that product coordinate.

For the last assertion, compare the inner columns and arrow columns at  $P_i = P_j = 0$ . The inner image is  $J(X_0) \otimes M_b$ , of dimension  $(d^2 - 1)b^2$ . The arrow derivative is

$$X_1 \otimes U_1 + X_2 \otimes U_2.$$

The original transverse independence implies that the combined map has rank  $(d^2 + 1)b^2$  at this reduction. Its generic rank is at least this number. At a generic pair, the inner map has rank at most  $(db)^2 = d^2b^2$ ; in fact it is injective, since a nonzero intertwiner between the two simple representations would identify their kernels. Subtracting this generic inner rank leaves at least  $b^2$  independent arrow directions in the quotient. A generic fixed arrow consequently has nonzero quotient class.  $\square$

All determinant conditions in Lemma 12.2 hold on the same independent block and arrow coordinates used in the division realization. Only finitely many blocks and arrows are involved, so their nonzero minors can be imposed simultaneously. In particular, the argument does not require replacing that realization by a different choice of data.

We next isolate the translation argument. Its hypotheses concern only the constant and linear coefficients; higher-order corrections may be arbitrary.

**Lemma 12.3** (A translation preserving two directions). *Let  $E$  be a field of characteristic zero, let  $n \geq 1$  be an integer, let  $m = 11$ , and index the blocks of  $M_{mn}(E)$  by  $\mathbb{Z}/m\mathbb{Z}$ . Suppose*

$$Y(s_1, s_2) \in M_{mn}(E) \langle\langle s_1, s_2 \rangle\rangle^u$$

*has constant tuple  $C = \mathrm{diag}(C_i)$  generating  $\prod_i M_n(E)$ . Suppose its two linear coefficients have the following supports:*

$$(Y_1)_{ij} = 0 \text{ unless } i - j \in \{1, 2\}, \quad (Y_2)_{ij} = 0 \text{ unless } i - j \in \{3, 4\}.$$

Assume every displayed edge coefficient is nonzero modulo the cross-inner image  $\{C_i T - T C_j : T \in M_n(E)\}$ . For a new central Laurent parameter  $\gamma$ , define

$$Y'(s'_1, s'_2) = Y(\gamma + s'_1, s'_2).$$

Then  $Y'_0 = Y(\gamma, 0)$  generates  $M_{mn}(E((\gamma)))$ , and  $Y'_1, Y'_2$  are independent modulo  $J(Y'_0)$ .

*Proof.* The substitution is well-defined. In the coefficient of a fixed new word, a contribution from an old word of length  $r$  contains  $r$  minus the new word length factors of  $\gamma$ . Thus only finitely many terms contribute at each  $\gamma$  degree. Coefficients converge in  $E[[\gamma]]$ , and the substitution respects products.

Let  $B = \prod_i M_n(E)$  and let  $I_C$  be the kernel of evaluation of the free algebra at  $C$ . First-letter differentiation on  $I_C$  has image a  $B$ -bimodule in  $M_{mn}(E)$ . Its projection onto every first-color edge is nonzero. Otherwise the derivative into that cross bimodule would descend to a derivation  $\delta : B \rightarrow M_n(E)$ , where the left and right actions are through the distinct endpoint factors  $i, j$ . Such a derivation is inner. In fact, for the central idempotent  $e_i$  of factor  $i$ , differentiating  $e_i a = a e_i$  gives

$$\delta(a) = a_i T - T a_j, \quad T = \delta(e_i).$$

Applied to the generators, this contradicts the transverse edge hypothesis. Since each full rectangular block is a simple bimodule for its two matrix factors, the derivative image contains the full block on every first-color edge.

Divide the corresponding evaluations of elements of  $I_C$  at  $Y(\gamma, 0)$  by  $\gamma$ . Their reductions supply all these edge blocks. Ordinary evaluations already supply the full diagonal product. The offset-one edges form a directed cycle through all vertices, so products supply every matrix block. A finite set of such elements with independent reductions therefore proves that  $Y(\gamma, 0)$  generates the full matrix algebra.

It remains to check the two directions. Put  $N = mn$  and  $J_0 = J(C)$ . Since  $C$  generates  $B$ , its commutant consists of the  $m$  block scalars. Hence  $\dim J_0 = N^2 - m$ . Choose  $N^2 - m$  inner columns with independent reductions spanning  $J_0$ . For a block scalar  $c = \text{diag}(c_i I_n)$ , the additional column

$$\gamma^{-1}[c, Y(\gamma, 0)]$$

is integral and reduces to  $[c, Y_1]$ . These columns have rank  $m - 1$  modulo  $J_0$ . Indeed,  $[c, Y_1] \in J_0$  forces  $c_i = c_j$  on each offset-one edge, by its transverse hypothesis, and the cycle forces all  $c_i$  to agree.

The two shifted linear coefficients reduce to  $Y_1, Y_2$ . Suppose

$$aY_1 + bY_2 \in J_0 + \{[c, Y_1] : c \text{ is block scalar}\}.$$

At a second-color edge, only  $bY_2$  contributes outside  $J_0$ , so  $b = 0$ . On the directed offset-one cycle, the relation then forces  $a = c_i - c_j$  on each edge. Summing gives  $ma = 0$ , whence  $a = 0$  in characteristic zero. Thus these two reductions are independent modulo the enlarged inner-column space.

We have found  $N^2 - m + (m - 1) + 2 = N^2 + 1$  independent reductions among inner columns and the two tangents. They lift to independent columns over  $E((\gamma))$ . Full generation of  $Y'_0$  implies  $\dim J(Y'_0) = N^2 - 1$ , so the two tangent classes are independent.  $\square$

Apply the lemma to  $Y = X(Z)$ . Lemma 12.2 supplies its diagonal-product and edge hypotheses, while the eleven-block construction supplies precisely the two color supports. Consequently the

translated tuple generates after splitting and has two independent tangent classes. These are finite rank conditions, so they also hold over the division algebra's own center. The central Laurent extension by  $\gamma$  preserves division. Both the old polynomial identities and the new algebraic relation are preserved by the convergent substitution.

## 12.1 Returning to a countable center

The complete Laurent fields used so far need not be countable. We finish by retaining only the scalar data required by the result. Use the notation of Section 11:  $F_0 = H_0(a_*)$  is the construction field,  $D_0$  is the central division algebra over it, and

$$L = F_0((\varepsilon))((\gamma))$$

is the full shifted Laurent center. The field  $F_0$  is countable:  $L_0$  is generated by finitely many indeterminates over the countable field  $k(\alpha, \beta)$ , and  $H_0$  is a subfield of  $L_0$ . The algebra  $D_0$  has degree  $dbm$ , and  $D_0 \otimes_{F_0} L$  is the central division algebra just constructed.

Choose an  $F_0$ -basis of  $D_0$ . Let  $k' \subseteq L$  be the field generated by  $F_0$ , the coefficients of the monic polynomial  $q$  (taken to be  $q_0$  when  $q_0(w) = 0$ ), and every scalar coordinate in this basis of every coefficient of the translated series  $X'$  and  $w'$ . There are countably many free words and finitely many coordinates per coefficient, so  $k'$  is countable. Put

$$A' = D_0 \otimes_{F_0} k'.$$

This is central simple of degree  $dbm$ . Its scalar extension to  $L$  is division, so  $A'$  embeds in a division ring and has no zero divisors. A finite-dimensional algebra without zero divisors is a division algebra. Thus  $A'/k'$  has the required central division property and strictly larger degree.

All coefficients of  $X', w'$  and  $q$  lie in this smaller model. The identities descend by its injective scalar extension to  $L$ . Generation and transverse independence descend as well, since the ranks of the finite coordinate matrices defining them do not change under field extension. This proves every conclusion of Theorem 2.1.

In the unconstrained case  $q_0(w) = 0$ , the same argument applies without the spectral corrections: use the same eleven-block generic constant and colored linear terms, retain the relation  $q_0$ , and carry out the translation and countable restriction above. Thus that case also increases degree while preserving generation and both directions.

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